Selective Reporting in Published Finance Research: A Two-Corpus Comparison of Government-Intervention Studies*

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This version: June 4, 2026

[Working paper — please do not cite without permission]

ABSTRACT

We apply the Brodeur et al. (2016) caliper test to 7,761 test statistics from 121 corporate-finance regulation articles in the *Journal of Finance*, *Review of Financial Studies*, and *Journal of Financial Economics* and compare them against 8,351 test statistics from 125 NBER working papers on the same topic. **Primary specification (verifiable rows):** among the 5,510 published rows with extractable coefficient-SE pairs, the caliper ratio is 0.81 — below one and statistically indistinguishable from the NBER ratio of 0.95. No systematic threshold bunching exists in primary regression rows of either corpus. **Secondary finding (standalone statistics):** published papers show elevated bunching in standalone z-statistics (ratio 2.55 vs. 1.49 for NBER), concentrated in supplementary and robustness tables (96.1% of published standalone rows come from non-primary tables). This pattern is consistent with selective reporting of which individual test results to highlight, rather than manipulation of primary regression estimates. The full-sample published ratio of 1.20 (cluster-bootstrap 95% CI [0.88, 1.66]; bootstrap $p = 0.150$; permutation $p = 0.001$) is approximately equally

*We thank seminar participants at [XX] for helpful comments. We are grateful to [XX] for outstanding research assistance. All data and code are available at https://github.com/[REPOSITORY].
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attributable to a within-type effect (standalone statistics bunching more in the published corpus, contributing +0.097 to the gap) and a composition effect (published papers contain more standalone-statistic rows: 31.6% vs. 21.0%, contributing +0.094). The NBER and journal samples are largely non-overlapping cross-sections (only ≈12% of published papers are matched to NBER counterparts), so this comparison reflects differences between two distinct corpora and cannot rule out compositional confounds.

A cross-sectional probit finds no detectable directional sorting: the binary directional coefficient is −0.020 (s.e. 0.182, $p = 0.91$, $N = 291$). This design rules out directional sorting ≥22 percentage points at 80% power; moderate asymmetries are consistent with the data. The strongest predictor of corpus membership is a quasi-experimental identification strategy (0.467***, s.e. 0.157), though this variable lacks a human-coded validation and likely captures a bundle of correlated quality attributes.

JEL classification: G18, G28, C80, B40.

Keywords: Publication bias; Government intervention; Finance journals; Identification quality; Caliper test; Working papers

1. Introduction

Academic research on government intervention in financial markets occupies a privileged position in public policy. The Federal Reserve Board, the Securities and Exchange Commission, and the Basel Committee on Banking Supervision routinely incorporate peer-reviewed finance research into regulatory impact assessments, rule-makings, and policy speeches. The Dodd-Frank Wall Street Reform and Consumer Protection Act alone generated an estimated 400 new regulatory mandates, many of which were designed in explicit reference to academic evidence on the effects of capital requirements, disclosure rules, and consumer financial protection. The credibility of this evidence base is therefore a first-order policy question: if the academic record systematically misrepresents which interventions have measurable effects—and which do not—regulators face a distorted picture that can generate welfare-reducing policy choices.

Publication bias offers a mechanism by which such distortion arises without any deliberate misconduct. When researchers are reluctant to write up null results, when journals are more likely to accept papers with statistically significant findings, or when both effects operate simultaneously, the published literature understates the true prevalence of null or inconclusive evidence (Franco et al., 2014). In a domain such as government intervention research, where policy decisions depend on whether a regulation has a measurable causal effect, suppression of null results generates the false impression that most interventions work—or, more precisely, that most interventions have discernible effects in one direction or the other.

Why might directional publication bias arise in the government-intervention literature specifically? Two theoretical channels operate in opposite directions, making the sign of any bias an open empirical question. A *pro-intervention channel* could arise if journals and readers favor results that confirm a policy's stated objective (e.g., that capital requirements reduce bank risk), both because positive results are more actionable for regulators and because they more naturally confirm the prior that motivated the policy. An *anti-intervention* *channel* could arise if research that documents unintended costs or failures of regulation is more surprising—and thus more publishable—than research confirming that a policy works as intended. Under either channel, the direction of sorting would differ from significance bias (null vs. directional), which reflects a simpler threshold between finding-something and finding-nothing. This paper is designed to detect both types.

This paper studies sorting patterns in a domain the existing literature has not examined: the finance literature's treatment of government intervention. Prior work on publication bias in finance has focused on the cross-section of asset pricing anomalies, documenting that factor-zoo proliferation reflects selection for high t-statistics (Harvey et al., 2016; Hou et al., 2020; Jensen et al., 2023). A parallel economics literature (Brodeur et al., 2016; Andrews and Kasy, 2019) infers bias from the shape of published p-value distributions, finding excess mass just below conventional significance thresholds. Neither strand addresses the policy-intervention literature, nor do they distinguish two conceptually distinct forms of bias with different welfare implications. *Significance bias*—directional findings over-published relative to null findings—makes every intervention appear detectable. *Directional bias*—pro-intervention findings over-published relative to anti-intervention findings—additionally skews the regulatory record toward a particular conclusion about the sign of policy effects. We design our study to test for both.

Our research design combines two complementary samples. The *published paper* sample consists of all in-scope articles published in the *Journal of Finance*, the *Review of Financial Studies*, and the *Journal of Financial Economics* from 2000 to 2024. The *working paper* sample is drawn from NBER working papers in five programs: Corporate Finance (CF), Asset Pricing (AP), Monetary Economics (ME), Economic Fluctuations and Growth (EFG), and Public Economics (PE), covering papers from 2000 to 2024. A key feature of this two-sample design is that we observe findings from working papers that circulated *before* potential journal publication, allowing a two-corpus comparison of test-statistic distributions rather than merely inferring the pre-publication shape from published p-value bunching alone. The NBER corpus serves as a proxy for the pre-publication distribution; it is selected on NBER affiliation and should not be equated with the full population of submitted or in-progress papers. We emphasize that the two samples are largely non-overlapping cross-sections (only $\approx$12% of published papers are matched to NBER counterparts), so the comparison reflects differences between two distinct corpora — not the same papers tracked through the editorial filter. The identifying assumption is that the NBER working-paper distribution is reasonably representative of research circulating before top-journal submission in this domain, conditional on NBER affiliation, which is itself selective. Together, these samples allow us to compare the full distribution of findings across publication states—not merely the subset that appears in top journals.

We define a paper as in-scope if its primary research question is the causal effect of a specific government law, regulation, or public policy on financial markets, firms, or investors. Identifying in-scope papers in a corpus of nearly 40,000 documents (39,136) would be infeasible by hand. We therefore develop a two-pass automated pipeline: a keyword filter that screens for government intervention terminology, followed by a large language model (LLM) classifier that reads each flagged abstract and returns a binary in-scope decision with a structured justification. For each in-scope paper we then code the *direction* of the primary finding as positive (+1, the intervention was beneficial), negative (-1, harmful), or null/mixed (0).

Our primary finding is the *absence* of a cross-sectional significance-sorting pattern in the published record. Corpus-membership rates are nearly identical across all three direction categories: positive-finding papers appear in top journals at 45.6%, negative-finding papers at 46.1%, and null-finding papers at 46.7%. A χ^2 test of homogeneity cannot reject equality ($\chi^2 = 0.017$, $p = 0.992$), and a two-sample z-test for equality of the positive and negative rates likewise fails to reject ($z = -0.07$, $p = 0.94$).

Probit models confirm the absence of a direction premium. The binary directional coefficient is -0.020 (s.e. 0.182, $p = 0.91$), and the three-way direction indicators are -0.026

(positive, $p = 0.90$) and -0.014 (negative, $p = 0.94$).[1] A Wald test for equality of the two direction coefficients gives $\chi^2 = 0.005$ ($p = 0.944$). These null results are robust across all six specifications.

The strongest predictor of corpus membership is the use of a quasi-experimental identification strategy. The *QuasiExp* dummy carries a coefficient of 0.467^{***} (s.e. 0.157, $p = 0.003$) in Column (6), implying an 18 percentage-point premium for papers using RDD, IV, DiD, or event-study designs. This premium survives all controls, whereas direction coefficients remain statistically indistinguishable from zero. This pattern is *consistent with* editorial selection operating on methodological quality rather than finding direction—though the LLM-coded quasi-experimental variable is likely correlated with author quality, institutional affiliation, and topic type, so the coefficient reflects a bundle of quality-correlated attributes and cannot be attributed to identification strategy alone.

The absence of a direction–corpus-membership association is uniform across journals; no individual journal shows a statistically significant direction coefficient (Appendix C), though sub-samples are small and cross-journal comparisons are severely underpowered.

A complementary analysis of the *within-paper* distribution of test statistics reveals a distinct pattern. We apply the Brodeur et al. (2016) caliper test to 7,761 $|z|$-statistics from 121 published papers, stratified by row type. *Primary specification (verifiable rows):* Among the 5,510 published rows with both a coefficient and a standard error, the caliper ratio is 0.81 — below one, with no bunching below the 5% threshold — and is statistically indistinguishable from the NBER ratio of 0.95. No threshold bunching exists in the systematically reported primary regressions of either corpus. *Secondary finding (standalone z-statistics):* In standalone statistics (rows reporting a $|z|$-value without a coefficient-SE pair, representing 31.6% of published rows), the published ratio is 2.55 versus 1.49 for NBER. The full-sample published ratio of 1.20 (cluster-bootstrap $p = 0.150$, CI $[0.88, 1.66]$; permutation

[1] The baseline implied publication probability at the bivariate intercept is $\Phi(-0.084) \approx 46.7\%$, nearly identical to all three direction-specific corpus-membership rates. Marginal effects for direction indicators are negligible throughout.

$p = 0.001$) reflects approximately equal contributions from a within-type effect (+0.097: standalone statistics bunch more intensely in published papers) and a composition effect (+0.094: published papers contain more standalone-statistic rows). The directional pattern for standalone statistics is imprecisely estimated (negative-finding published: ratio 1.52, cluster-bootstrap $p = 0.135$, CI $[0.81, 2.67]$); two of three paper-level inference procedures are insignificant, and the finding should be treated as suggestive.

Sub-period estimates across three regulatory regimes (2000–2007, 2008–2015, 2016–2024) are uniformly insignificant, with all direction coefficients well below conventional significance thresholds. Sub-period sample sizes ($N = 50$, 95, 146) are small and these results should be interpreted as descriptive, not confirmatory.

This paper makes three contributions.

First and primarily, it documents a *pattern of selective reporting* in the within-paper distribution of test statistics in published government-intervention finance research. Specifically: systematically reported primary regression rows (with verifiable coefficient-SE pairs) show no caliper bunching and no gap relative to NBER working papers. Standalone z-statistics—predominantly from supplementary and robustness tables—show elevated bunching in published papers (ratio 2.55) relative to NBER (ratio 1.49), a difference that is suggestive but imprecisely estimated (cluster-bootstrap $p = 0.150$). This pattern is consistent with authors selectively reporting which supplementary test statistics to highlight, rather than with manipulation of primary regression estimates. Using NBER full texts as a proxy for the pre-publication distribution, we provide a two-corpus comparison that avoids the purely inferential step required when the pre-publication distribution is unobserved; the two corpora are largely non-overlapping, so compositional confounds cannot be ruled out.

Second, as supporting context, the paper provides cross-sectional evidence consistent with the absence of detectable *directional* sorting between top-journal articles and NBER working papers. This null result rules out effects $\geq$22 percentage points at 80% power; moderate asymmetries remain consistent with the data and cannot be distinguished from zero at the current sample size.

Third, the paper introduces a scalable LLM-assisted pipeline for classifying and direction-coding large economics paper corpora. Throughout, we are explicit that our design identifies cross-sectional associations, not causal editorial-filter effects.

The remainder of the paper proceeds as follows. Section 2 reviews the related literature. Section 3 describes data construction and the classification pipeline. Section 4 presents the empirical strategy. Section 5 reports main results and economic magnitudes. Section 6 presents robustness checks. Section 7 discusses mechanisms, policy implications, comparisons to other domains, and limitations. Section 8 concludes.

2. Related Literature

2.1 Publication Bias in Asset Pricing

The closest prior work on publication bias in finance examines the cross-section of expected returns. Harvey et al. (2016) survey hundreds of return predictors published in top finance journals and argue that the conventional $t > 2$ threshold is inadequate given the scale of multiple testing in the factor zoo. Hou et al. (2020) attempt to replicate 452 published anomalies and find that 65% fail under standard portfolio construction choices, consistent with upward-biased published effect sizes. Jensen et al. (2023) document that most factors replicate out-of-sample but that the factor space is highly redundant, implying substantial duplication in published claims. Harvey and Liu (2021) develop a structural model to infer the pre-publication distribution of test statistics—the full iceberg—from the published distribution alone, which is only the tip. They estimate the share of true factors and the intensity of specification search as latent parameters; identification is challenging because the pre-publication distribution is unobserved. Chen (2021) takes a complementary approach, running thought experiments to bound how much p-hacking would be needed to generate the observed right tail of published t-statistics if all 300+ asset-pricing factors were spurious.

He finds the implied search effort is implausibly large—if 10,000 researchers generated one factor per minute it would take millions of years—concluding that most published factors reflect genuine economic content rather than data-mining. Both papers study the *level* of published test statistics in asset pricing.

This strand focuses entirely on *significance-level* sorting in the asset-pricing factor zoo. All papers in it face the identification problem that the pre-publication distribution is unobserved and must be inferred from a structural model (Harvey & Liu) or bounded by thought experiment (Chen). Our paper sidesteps this problem by using NBER working paper full texts as a proxy for the pre-publication distribution, converting an otherwise purely inferential exercise into a direct two-corpus comparison—though the NBER corpus is itself selected on researcher affiliation and is not equivalent to the full pre-publication distribution.

The two papers are also studying a different part of the t-statistic distribution. Chen's bound concerns the *far* right tail ($|t| > 4$, $|t| > 6$), which is too heavy to be explained by p-hacking alone. Our caliper test focuses on the narrow window *around* the 5% threshold ($|z| \in (1.645, 2.275)$), where selective reporting of marginally significant specifications creates excess mass just *below* the 1.96 boundary. Both phenomena can coexist: genuine economic effects produce the fat tail; the caliper bunching concentrated in the published record is consistent with revision-stage specification pressure producing excess mass in the 10%-significant-but-not-5% zone. Indeed, our findings are entirely consistent with Chen's conclusion that most published effects are real—we find no evidence that publication *directionally* sorts pro- versus anti-intervention findings, and our caliper evidence is consistent with within-paper revision pressure rather than outright fabrication.

Our paper also differs on the question asked. Factor-return sign is not normatively coded, so these papers cannot test for *directional* bias—whether papers concluding an intervention is harmful are less likely to reach print than papers concluding it is beneficial. That question is economically meaningful only in a domain where finding direction has a clear policy interpretation, which is precisely the government-intervention literature we study.

2.2 Publication Bias in Economics

The economics literature on publication bias draws on a long tradition in statistics and medicine. Brodeur et al. (2016) analyze 50,000 hypothesis tests reported in the *American Economic Review*, the *Journal of Political Economy*, and the *Quarterly Journal of Economics*, finding excess mass just below the 5% significance threshold ($|z| \in (1.645, 1.96)$), consistent with selective reporting of marginally significant specifications. Andrews and Kasy (2019) provide a structural estimator for publication bias that allows recovery of the bias-corrected effect-size distribution from the published sample. Ioannidis et al. (2017) document systematic underpowering across economics studies, which mechanically amplifies bias whenever only significant findings are reported. Christensen and Miguel (2018) survey the broader reproducibility literature and document that publication bias, p-hacking, and HARKing (hypothesizing after results are known) are pervasive in empirical economics. These approaches infer bias from the internal shape of the published distribution—they do not observe pre-publication findings and therefore cannot attribute the sorting gap to editorial selection specifically. Our paper contributes domain-specific evidence for the finance-regulation literature using a pre-publication baseline that identifies the editorial stage directly.

2.3 Government Intervention in Financial Markets

Academic finance contains a large and consequential body of work on the effects of securities law, banking regulation, and macroprudential policy. Studies of financial regulation (Berger and Bouwman, 2017), law and finance (La Porta et al., 1998), insider trading enforcement (Bhattacharya and Daouk, 2002), short-selling restrictions (Bris et al., 2007), and the Dodd-Frank Act (Agarwal et al., 2014) have produced heterogeneous findings across interventions, time periods, and countries. Whether the published distribution of findings faithfully represents the underlying distribution of true effects has not previously been addressed for this literature. We fill this gap.

2.4 LLMs in Economics Research

Recent work demonstrates that large language models can accurately classify economic texts, extract structured data from unstructured documents, and replicate tasks that previously required extensive human coding (Ash and Hansen, 2023; Korinek, 2023). We contribute a validated pipeline for in-scope classification and direction coding of finance abstracts, along with a publicly released coding rubric and prompt. Our two-pass design (keyword pre-filter plus LLM classification) reduces API costs and allows replication-based validation. We conduct an independent second coding run on a 60-paper stratified subsample, obtaining a linear-weighted Cohen's kappa of $\kappa = 0.674$, consistent with "substantial agreement" by the Landis and Koch (1977) classification, and addressing concerns about LLM reliability in structured classification tasks (Brodeur et al., 2023).

3. Data and Sample Construction

3.1 Published Papers

The published-paper sample consists of all articles appearing in the *Journal of Finance*, the *Review of Financial Studies*, and the *Journal of Financial Economics* from 2000 to 2024. These three journals constitute the traditional top tier of academic finance and account for the majority of highly cited finance research on government intervention. We collect article metadata—title, authors, year, volume, issue, DOI, and abstract—for all articles in each journal over this period.

Journal articles are identified and retrieved as follows. For the *Journal of Finance* and the *Review of Financial Studies* we use the Web of Science (WoS) full-record download, supplemented by publisher APIs where abstract text is missing. For the *Journal of Financial Economics*, abstracts were obtained via the Elsevier ScienceDirect Article Retrieval API, which provides metadata including the `dc:description` field for all Elsevier-hosted articles.

Table 1 summarizes the journal sources.

Table 1. Journal Sample

Journal	Abbrev.	ISSN	Publisher	Abstract source
Journal of Finance	JF	0022-1082	Wiley	Web of Science
Review of Financial Studies	RFS	0893-9454	Oxford	Web of Science
Journal of Financial Economics	JFE	0304-405X	Elsevier	ScienceDirect API

Coverage: 2000–2024. Abstract text is available for 100% of *Journal of Finance* research articles, 100% of *Review of Financial Studies* articles, and 100% of *Journal of Financial Economics* articles after manual abstract enrichment. The *Journal of Finance* denominator is restricted to research articles only; the raw Web of Science record for the *Journal of Finance* includes a large number of editorial matter, meeting announcements, and AFA administrative items (which carry no abstracts), so the overall entry-level coverage rate is 74.6%. Our pipeline excludes these non-article entries before computing coverage. Because the pipeline classifies NBER working papers using NBER abstracts—not journal abstracts—any residual *Journal of Finance* coverage gap does not affect our classification results.

3.2 NBER Working Papers

The working-paper sample is drawn from the NBER Working Paper Series via the public API (`https://api.nber.org`). We collect all papers published from 2000 to 2024 under five programs: Corporate Finance (CF), Asset Pricing (AP), Monetary Economics (ME), Economic Fluctuations and Growth (EFG), and Public Economics (PE). These programs collectively encompass the dominant share of academic finance research on government intervention. Papers appearing in multiple programs are counted once.

We use the NBER working paper sample as a reference distribution for comparison with the published sample. The identifying assumption required for a causal interpretation is that NBER working papers represent a reasonably broad cross-section of pre-submission findings in this domain—one not itself selected on the direction of findings. We note this assumption is strong: NBER affiliation is selective on researcher quality and institutional prestige, and many finance papers circulate through SSRN rather than NBER. The NBER distribution may therefore differ from the full pre-publication distribution for reasons unrelated to editorial selection. We return to this identification challenge formally in Section 4.1.

3.3 In-Scope Classification Pipeline

Identifying government-intervention papers in a corpus of 39,136 documents requires an automated approach. We apply a two-pass pipeline designed to maximize recall while maintaining precision. Table 2 reports the number of papers surviving each stage of the selection process.

Starting universe. We begin with all articles and working papers collected from our four sources—the *Journal of Finance*, the *Review of Financial Studies*, the *Journal of Financial Economics*, and the NBER Working Paper Series—for the period 2000–2024, yielding a combined corpus of 39,136 documents (30,212 NBER working papers plus 8,924 journal articles: 3,383 from the *Journal of Finance*, 2,497 from the *Review of Financial Studies*, and 3,044 from the *Journal of Financial Economics*).

Pass 1 — Keyword filter. A paper passes the first filter if the combined text of its title and abstract contains at least one term from a pre-specified taxonomy of government intervention keywords spanning seven categories: financial regulation (e.g., "Dodd-Frank," "capital requirements," "Basel"), securities law (e.g., "SEC," "Sarbanes-Oxley," "insider trading law"), banking policy (e.g., "FDIC," "deposit insurance," "TARP"), monetary and fiscal policy (e.g., "quantitative easing," "fiscal stimulus"), market microstructure rules (e.g., "uptick rule," "tick size pilot"), corporate governance law (e.g., "say-on-pay," "proxy access," "mandatory disclosure"), and general intervention signals (e.g., "government intervention," "public policy," "law and finance"). Ambiguous standalone terms require co-occurrence with a policy-specific second keyword. The full taxonomy is reported in Appendix A.

Searching the combined title-and-abstract text rather than the abstract alone improves recall for three reasons. First, authors frequently name the specific law or regulation in the title but refer to it only as "the reform" or "the rule" throughout the abstract, so the keyword would appear only in the title. Second, common abbreviations (e.g., "Dodd-Frank" in the title) and full statutory names (e.g., "Wall Street Reform and Consumer Protection Act" in the abstract) are not always used interchangeably; searching both fields catches either form. Third, because false positives are inexpensive to discard in Pass 2 whereas false negatives are permanent, Pass 1 is deliberately over-inclusive: the cost of an extra LLM call is negligible relative to the cost of omitting a genuine in-scope paper.

The keyword filter reduces the corpus from 39,136 documents to **1,049 candidates** (510 NBER, 184 *Journal of Finance*, 151 *Review of Financial Studies*, and 204 *Journal of Financial Economics*).

Pass 2 — LLM classifier. Each keyword-flagged candidate is passed to Claude (claude-opus-4-7; Anthropic 2024) with the following instruction (full prompt in Appendix B):

> *"A finance research paper is 'in scope' if and only if its PRIMARY research question is the CAUSAL EFFECT of a specific government law, regulation, or public policy on financial markets, firms, or investors. Purely theoretical papers, papers that only describe a regulation without estimating its effect, and papers whose main topic is private firm behavior with policy as incidental context are excluded."*

The model returns a JSON object with three fields: `in_scope` (boolean), `confidence` (high, medium, or low), and a one-sentence `reason`. We run the classifier with eight parallel threads to maximize throughput; rate-limit errors are handled via exponential-backoff retries.

Following the automated LLM pass, we apply multiple rounds of manual validation to enforce the causal-effect criterion more strictly. The most consequential step removes papers that use a regulatory event *only* as an exogenous instrument to identify a non-regulatory relationship—for example, using SOX passage to instrument for disclosure quality in order to study its effect on firm investment. Such papers' primary research question is not the policy effect itself; they fall under the third LLM exclusion criterion (main topic is private behavior with policy as incidental context) but were not consistently flagged by the automated classifier. We subsequently re-ran Pass 1 using the full NBER abstracts (retrieved from nber.org rather than the truncated API text of ~45 words), which yielded 803 additional NBER candidates. The automated pipeline and manual validation together retain **343 in-scope papers** from the combined 1,852 candidates (1,049 initial plus 803 from the full-abstract re-run). Two papers whose primary focus is exchange-imposed market microstructure rules rather than government policy are then excluded. The final analysis sample consists of **291 papers**: 157 NBER working papers and 134 published journal articles (26 in the *Journal of Finance*, 48 in the *Review of Financial Studies*, and 60 in the *Journal of Financial Economics*).

Table 2. Sample Selection

Selection step	Papers remaining	Papers dropped
Starting universe (2000–2024)		
NBER working papers	30,212	
Journal articles: JF	3,383	
Journal articles: RFS	2,497	
Journal articles: JFE	3,044	
Total corpus	**39,136**	
Pass 1: Keyword filter		
Retain keyword-hit papers	1,049	38,087
of which: NBER	510	
of which: JF	184	
of which: RFS	151	
of which: JFE	204	
Re-run Pass 1 with full NBER abstracts	+803 new candidates	
Pass 2: LLM scope classification		
Retain in-scope papers (LLM, combined)	343	1,509
Exclude exchange-imposed market structure rules	341	2
Manual validation and scope review	291	50
Final analysis sample		
Published in JF / RFS / JFE	134	
of which: JF	26	
of which: RFS	48	
of which: JFE	60	
NBER working papers (unpublished)	157	
Total	**291**	

Notes. The starting universe consists of all NBER working papers under programs CF, AP, ME, EFG, and PE from 2000 to 2024, plus all research articles retrieved from the *Journal of Finance*, *Review of Financial Studies*, and *Journal of Financial Economics* over the same period. Pass 1 applies a keyword taxonomy of government intervention terminology (Appendix A); Pass 2 uses a large language model (Claude) to verify that each flagged paper's primary research question is the causal effect of a specific government policy on financial markets, firms, or investors. The 341 papers surviving the automated exclusion steps comprise both journal articles (Published = 1) and NBER working papers. The subsequent manual validation and scope review removes 50 papers: 16 in-scope NBER working papers that are matched to a top-journal publication (represented by their published version) and 34 NBER papers reclassified as out-of-scope or theoretical during manual review. The final 291 analysis observations therefore consist of 134 published journal articles (Published = 1) and 157 NBER working papers with no confirmed journal match (Published = 0). The published sub-sample comprises all in-scope journal articles regardless of whether a corresponding NBER working paper exists.

3.4 Direction Coding

For each in-scope paper, we assign a direction code to the paper's primary empirical finding: $+1$ (the government intervention produced a positive or beneficial outcome), -1 (negative or harmful), or 0 (null, mixed, or inconclusive). Papers with multiple interventions are coded on the primary intervention identified from the abstract. Direction coding uses the same model as the in-scope classifier (claude-opus-4-7, model ID: `claude-opus-4-7`, accessed June 2025; Anthropic 2024). Because closed-weight models may produce different outputs as they are updated over time, this is a genuine threat to exact replication; we therefore provide all input abstracts and output codes in the replication package alongside the exact model IDs used.

The coding prompt instructs the model to code $+1$ if the intervention produced a *positive or beneficial outcome relative to its stated regulatory objective*, -1 if harmful or counterproductive, and 0 if null or mixed. Normative framing by the authors is explicitly excluded. This distinction matters: a paper finding that capital requirements reduce bank risk-taking is coded positive ($+1$) because the intervention achieved its intended regulatory goal, regardless of whether the authors believe capital requirements are optimal policy. The full coding prompt is in Appendix B.

Coding reliability. We validate the direction codes through two exercises.

Model-choice sensitivity check (inter-run reliability). We re-ran the direction coding protocol on a stratified random sample of 60 papers (5 papers per source $\times$ direction cell, balanced across the three journals and NBER), treating the independent replication run as a second rater. The replication run used a smaller model (claude-haiku-4-5-20251001, temperature = 0) with the same prompt; disagreements therefore isolate model-choice sensitivity rather than prompt-choice sensitivity. We emphasize that two LLM runs sharing a prompt are *not* independent raters in the same sense as two trained human coders, and the reported κ should be interpreted as a model-stability check, not as true inter-rater agreement.

Prompt-perturbation robustness (systematically varying instruction wording) is a natural extension we leave for future work. The two runs agree on 43 of 60 papers (71.7%). The linear-weighted Cohen's κ across all three direction categories is $\boldsymbol{\kappa = 0.674}$, consistent with "substantial agreement" by the Landis and Koch (1977) scale ($\kappa \in [0.61, 0.80]$). Agreement is near-perfect for directional papers: the negative (-1) category achieves 100% agreement (20/20) and the positive ($+1$) category 90% (18/20). Disagreements are concentrated in the null/mixed category (25% agreement, 5/20); all 15 disagreements are null-to-directional flips, never directional-to-null. This asymmetry implies that any coding error inflates our estimated directional premium, making our reported estimates a conservative upper bound on the true premium. Because the null/mixed category is the probit omitted category, the joint χ^2 equality test ($p = 0.992$) is the most robust probit result, as it does not depend on the direction of the contrast against the null baseline. Appendix Table A4 reports the full confusion matrix; Appendix H disaggregates agreement by source corpus.

Human validation (pending). To assess agreement between LLM codes and a human gold standard, we drew a stratified random sample of 40 papers (20 published, 20 NBER) and coded each paper's direction from the abstract without reference to the LLM output. The coding sheet and comparison script (`scripts/13_human_validation_kappa.py`) are included in the replication package. A completed human-vs-LLM κ for this sample is a key remaining validation step; until it is available, the model-choice sensitivity $\kappa = 0.674$ above is the paper's primary reliability evidence.

3.5 Linking NBER Papers to Published Papers

Our probit analysis requires knowing which NBER working papers were eventually published in one of the three journals. We identify these links through a four-round pipeline in which each round searches only among the papers left unmatched by earlier rounds.

In the *first round* we apply token-sort Levenshtein similarity (RapidFuzz) to normalized titles, comparing each of the 134 journal papers against the full NBER archive of 30,212 working papers. A shared author surname is required as a second gate. Scores of 90 or above are automatically linked; scores in $[75, 90)$ are flagged for our manual inspection. In the *second round* we query six external sources—OpenAlex, Semantic Scholar, the NBER search endpoint, CrossRef (queried in both the preprint-to-journal and journal-to-preprint directions), and Unpaywall—and complement these with TF-IDF cosine similarity over paper abstracts with an author-overlap gate. In the *third round* a large language model (Claude) reads each unmatched journal paper's abstract alongside the first four pages of the top-40 NBER candidate PDFs and decides whether they describe the same research project. In the *fourth round* we supply Claude with the complete published journal PDF alongside the NBER draft, enabling identification of matches where both the title and the abstract were substantially rewritten during peer review.

The pipeline identifies 16 matched pairs among the 134 in-scope journal articles. The majority of matches are identified in Round 1 via fuzzy title matching; Rounds 2–4 contribute the remainder. The remaining 118 journal articles (88.1%) are unmatched. This reflects the prevalence of SSRN and institutional working papers in corporate-finance research rather than any limitation of the matching pipeline, which by Round 4 has exhausted all feasible linking methods including full-text comparison of published and preprint versions. External verification via OpenAlex, Unpaywall, and CrossRef independently confirms that these 118 papers have no retrievable NBER preprint record.

To assess whether the two samples are comparable in their directional composition, we compare direction rates across the full NBER and published sub-samples. Among all 157 NBER working papers, 79.6% have a directional finding; among the 134 directly identified journal articles, 79.1% do. These rates are nearly identical, suggesting that the two populations are broadly comparable in their directional mix. The close alignment in direction rates foreshadows the main finding: corpus-membership rates across directions are essentially equal.

In our probit analysis, unmatched journal papers are treated as having been published in a top journal (Published = 1), while NBER working papers without a confirmed journal match are treated as not yet published (Published = 0). This design identifies the probability that an in-scope paper appears in the top three journals as a function of its direction code.

3.6 Summary Statistics

Table 3 presents summary statistics for the 291-paper analysis sample. The direction distribution is strikingly similar between the NBER and published sub-samples. Among NBER working papers, 20.4% are coded null or mixed, compared with 20.9% among published papers—a difference of less than 1 percentage point. Positive-finding papers constitute 35.1% and negative-finding papers 44.0% of the published sample, versus 35.7% and 43.9% of the NBER sample. These near-identical distributions foreshadow the main result: no direction-based sorting at the publication stage.

Table 3. Summary Statistics

Sample	N	Positive (%)	Negative (%)	Null (%)	Mean year
All papers	291	35.4	44.0	20.6	2014.4
NBER working papers	157	35.7	43.9	20.4	2013.5
Published (JF/RFS/JFE)	134	35.1	44.0	20.9	2015.6
By journal:					
Journal of Finance	26	42.3	46.2	11.5	2012.7
Review of Financial Studies	48	29.2	54.2	16.7	2017.1
Journal of Financial Economics	60	36.7	35.0	28.3	2015.7

Notes. The sample consists of 157 NBER working papers (programs CF, AP, ME, EFG, and PE, 2000–2024) and 134 published journal articles (*Journal of Finance*, *Review of Financial Studies*, *Journal of Financial Economics*) classified as in-scope government-intervention papers by our LLM pipeline and manual validation. Direction codes: +1 = positive/beneficial finding, −1 = negative/harmful finding, 0 = null or mixed finding. The NBER and Published samples are mutually exclusive: the 291 papers consist of 157 distinct NBER working papers (unpublished) and 134 distinct published journal articles, with no paper appearing in both groups. The Published count of 134 is what enters the probit regressions as Published = 1. Column percentages may not sum to 100 due to rounding.

4. Empirical Strategy

4.1 Research Design and Identification

Before presenting the regression specification, it is important to be explicit about what our design does and does not identify. Our empirical strategy is a *cross-sectional comparison*: we observe a sample of papers that were circulated as NBER working papers and a separate sample of papers that were published in the *Journal of Finance*, the *Review of Financial Studies*, or the *Journal of Financial Economics*, and we ask whether the direction distribution of findings differs between the two groups. This is *not* a within-paper design in which the same papers are tracked from submission to publication outcome. Only 16 of 134 published papers (11.9%) are matched to an NBER working paper, so the two samples are largely non-overlapping populations.

The identifying assumption is that NBER working papers constitute a draw from the pre-publication distribution of findings in this domain of academic finance, conditional on NBER affiliation. Under this assumption, systematic differences in the direction distribution between the NBER and published sub-samples are informative about the editorial filter. This assumption is analogous to the design of Franco et al. (2014), who use NSF pre-registration to observe the pre-publication distribution of social-science findings; and to DellaVigna and Linos (2022), who obtain institutional RCT records. Those stronger designs are not available in corporate-finance research: there is no mandatory pre-registration requirement for empirical finance papers, no institutional repository that captures a broad cross-section of in-progress work, and no NSF-equivalent funding requirement that would force pre-registration. We use the NBER series as the closest available proxy for the pre-publication distribution.

Our design is weaker because NBER affiliation is selective on researcher quality and institution, and because most published finance papers circulate through SSRN rather than NBER. The direction of the resulting bias is ambiguous: NBER selection could make estimates *conservative* (if high-quality NBER researchers more often produce genuine null results, our null-paper baseline is unusually good, and the true gap for average researchers is larger); or it could make estimates *overstated* (if productive NBER researchers circulate work only once it is results-ready, the NBER corpus may be upwardly selected toward directional findings relative to all ongoing research, inflating the measured gap). We cannot resolve this ambiguity without submission-level data. We are explicit throughout that the coefficients in our probit models are cross-sectional associations, not causal publication-filter effects.

Critical design limitation: the two samples are largely non-overlapping populations, not the same papers at two pipeline stages. Because only 11.9% of published papers are matched to an NBER counterpart, the NBER and journal samples are overwhelmingly different papers selected by different institutional processes. This is the most fundamental constraint on the cross-sectional analysis. The constructed corpus-membership rate of 46% is a mechanical ratio determined by our sample construction—not an estimate of true journal acceptance probabilities. It depends on NBER program coverage, abstract availability, and the decision to count unmatched NBER papers as unpublished. We do not observe papers transitioning through the editorial filter; we compare the *direction distribution* of one corpus (NBER working papers) to that of another (published articles), and ask whether the two distributions are consistent with a significance-sorting mechanism.

The NBER benchmark is also imperfect in multiple ways beyond the matching rate. NBER affiliates are not representative finance researchers: they differ systematically in institutional quality, co-author networks, and publication success. Topic selection within NBER programs emphasizes certain areas, excluding papers that circulate through SSRN, CEPR, ECGI, or other channels. Timing selection is also a concern: researchers may circulate NBER working papers only after obtaining strong results, which would mean the NBER corpus is already partially selected toward directional findings relative to all ongoing research. These concerns cannot be fully resolved without submission-level data. We are explicit throughout that the probit coefficients are cross-sectional associations in this sample, not causal estimates of editorial selection effects. The key identifying assumption—that the NBER distribution is representative of pre-publication research in this domain conditional on NBER affiliation—is maintained for tractability, and we acknowledge it may not hold.

4.2 Primary Specification

Our primary test estimates a probit model at the paper level:

$$\Pr(\text{Published}_i = 1) = \Phi\big(\alpha + \beta_1 \text{Positive}_i + \beta_2 \text{Negative}_i + \boldsymbol{\gamma}'\mathbf{X}_i\big)\,, \tag{1}$$

where $\Phi(\cdot)$ is the standard normal CDF, Published_i equals one if in-scope paper i appears as an article in the *Journal of Finance*, the *Review of Financial Studies*, or the *Journal of Financial Economics*, Positive_i (Negative_i) is an indicator equal to one if the LLM-assigned direction code is +1 (-1), and the omitted category is papers with direction code 0 (null or mixed). The vector $\mathbf{X}_i$ includes a linear year trend and, in extended specifications, decade fixed effects.

Our primary coefficients of interest are β_1 and β_2. A finding of $\beta_1 > 0$ ($\beta_2 > 0$) indicates that positive-finding (negative-finding) papers are more likely to be in the published sample than null-finding papers, conditional on controls. Importantly, the probit coefficient $\hat{\beta}$ does not estimate an acceptance probability: it measures the conditional correlation between direction code and corpus membership across two non-overlapping cross-sections, not the probability that any individual paper passes through the editorial filter. A test of $H_0\colon \beta_1 = \beta_2$ addresses the directional-bias hypothesis: rejection would indicate that the publication premium depends on the sign of the finding, not merely on statistical significance.

4.3 Marginal Effects and Economic Interpretation

Because the probit model is nonlinear, we report marginal effects evaluated at the null-paper baseline:

$$\Delta_d = \Phi(\hat{\alpha} + \hat{\beta}_d) - \Phi(\hat{\alpha}), \quad d \in \{+, -\}.$$

This equals the difference in fitted publication probabilities between a directional-finding paper and a null-finding paper. We also report raw corpus-membership rates directly from the data as a transparent, model-free summary of the unconditional patterns.

4.4 Cross-Journal and Time-Trend Analysis

We re-estimate equation (1) separately for each journal to test whether the significance-sorting pattern is concentrated in one outlet or distributed uniformly across the top three. For the time-trend analysis we split the sample into three sub-periods corresponding to distinct regulatory regimes: 2000–2007 (pre-crisis baseline), 2008–2015 (financial crisis and post-crisis regulatory response), and 2016–2024 (post-Dodd-Frank implementation). We report results for all three sub-periods with appropriate caveats about sample size for the earliest period ($N = 50$).

5. Results

We organize results caliper-first. Section 5.1 presents the primary within-paper test-statistic analysis; Section 5.2 presents the cross-sectional direction-sorting analysis as supporting context.

5.1 Within-Paper Test Statistic Distributions

We begin with the paper's primary empirical test. The analysis proceeds in two layers. The *primary specification* restricts to rows with verifiable coefficient-SE pairs—rows where $|z| = |\hat{\beta}/\hat{\sigma}|$ can be independently confirmed. The *secondary analysis* examines standalone z-statistics (rows reporting a $|z|$-value without a coefficient-SE pair), which account for the majority of the full-sample bunching. We apply the Brodeur et al. (2016) caliper test to the full distribution of test statistics extracted from published and NBER paper PDFs.

Data. We extract all regression-table test statistics from the PDFs of the 134 published papers using `pdfplumber` for text extraction and a large language model to parse coefficient–

standard-error pairs into absolute z-values. After removing summary-statistics tables, intercepts, and fixed-effect indicators, the *published* sample contains **7,761 test statistics** from **121 papers** (JF: 1,308; RFS: 2,581; JFE: 3,872); the remaining 13 papers are pure theory models or have image-based tables. We apply the identical pipeline to the full-text PDFs of the 157 in-scope NBER working papers, obtaining **8,351 test statistics** from **125 papers** (79.6% paper coverage). The remaining 32 papers yield no extractable regression statistics; these are predominantly pure theory models, structural calibration exercises, or papers using non-standard empirical formats (synthetic control, non-parametric tests) without standard coefficient–standard-error tables. A small number (≤ 5 per corpus) appear to use image-embedded tables that `pdfplumber` cannot parse as text; these are treated as zero-stat and excluded from caliper analysis identically to the theoretical papers. The *NBER sample* serves as a proxy for the pre-publication distribution of test statistics from the same research domain—the closest available approximation in corporate-finance research, though it is selected on NBER affiliation and is not equivalent to the full pre-publication distribution. This allows a two-corpus comparison with the published distribution, rather than purely inferring bias from the shape of the published distribution alone.

Extraction spot-check. To validate extraction accuracy, we drew a random spot-check sample of 50 (paper, statistic) pairs—25 from the published corpus and 25 from NBER—and recorded the extracted coefficient, standard error, and implied $|z|$-value for each row. The spot-check sample (with blank "verified" and "PDF $|z|$" columns for manual comparison) is provided in `output/tables/extraction_spotcheck_sample.csv` in the replication package. Across-paper statistics: average $|z|$-value in the published corpus is 3.41 with a median of 2.84, consistent with high significance rates in published finance research; the NBER distribution is similar (mean 3.32, median 2.71). Within-paper counts range from 2 to 107 (published) and 1 to 244 (NBER), with medians of 71 and 86 respectively. These counts are broadly consistent with the number of regression rows one would expect to extract from a 40–60 page empirical finance paper containing 5–8 tables of 8–15 coefficients each.

Extraction quality by corpus. A potential concern is that differential extraction quality between typeset publisher PDFs (published) and NBER draft PDFs could itself generate the published-vs.-NBER caliper gap. We assess this in three ways. First, internal consistency: for statistics where we extract both a coefficient and a standard error, we verify that the reported $|z|$-value matches $|\hat{\beta}/\hat{\sigma}_{\hat{\beta}}|$ to within 0.1. This internal check passes for **98.3%** of published rows (5,474/5,569) and **99.7%** of NBER rows (6,811/6,831), giving no evidence of materially worse extraction quality in either corpus. The higher share of z-stat-only rows in the published corpus (29.0% vs. 18.7% for NBER) reflects the typeset formatting of some publisher tables that display t-statistics directly rather than coefficient-SE pairs; these statistics are taken at face value and are not subject to a consistency check, but they pass all caliper-window bounds checks. Second, the 13 published papers and 32 NBER papers that yield zero extracted statistics are distributed uniformly across journals ($\chi^2 = 1.08$, $p = 0.58$ for published) and by vintage (Mann-Whitney $p = 0.749$; median year 2017 for both kept and dropped papers), providing no evidence of systematic non-random exclusion. Third, restricting the caliper analysis to high-coverage papers ($\geq$20 extracted statistics per paper) leaves the published ratio essentially unchanged (1.207 vs. 1.203 for the full sample) and the NBER ratio unaffected (1.012 vs. 1.012). Fourth, we stratify the caliper by row type (Appendix K; summarized in Table A8). *Primary specification (coef+SE rows):* among the 5,510 published rows with both a coefficient and a standard error, the caliper ratio is **0.81** (below = 320, above = 394; naive $p = 0.006$); the corresponding NBER ratio is **0.95** (below = 416, above = 438; $p = 0.45$). Neither corpus shows excess bunching below the 5% threshold in verifiable rows, and the published ratio (0.81) is below the NBER ratio (0.95), not above it. This is the paper's primary caliper result: no threshold bunching in systematically reported primary regressions. *Secondary finding (standalone z-statistics):* among the 2,251 published z-stat-only rows, the caliper ratio is 2.55 ($p < 0.001$); the comparable NBER ratio is 1.49 ($p = 0.019$). Both corpora show bunching in standalone statistics, with published papers showing a substantially higher ratio.

To assess how much of the full-sample gap (published 1.20 $-$ NBER 1.01 = +0.191) is driven by composition versus within-type effects, we decompose:

- *Within-type effect* (published bunches more per se in standalone rows, evaluated at NBER's z-stat-only share of 21.0%): +0.097
- *Composition effect* (published has more z-stat-only rows: 31.6% vs. 21.0%): +0.094

The two effects are nearly equal in magnitude. Even if the composition effect is entirely explained by typesetting differences—published journals more often display t-statistics inline—roughly half the full-sample gap (+0.097) remains as a within-type effect: standalone z-statistics bunch more in published papers than in NBER papers of the same type.

Star-consistency validation of z-stat-only rows. To address whether the bunching in z-stat-only rows reflects extraction artifacts rather than real statistics, we cross-validate each z-stat-only row's extracted $|z|$ value against its reported significance stars. Under the standard convention (*** = 1%, ** = 5%, * = 10%), a row with extracted $|z| \in (1.645, 1.96)$ should carry a "*" star; a row with $|z| \in (1.96, 2.576)$ should carry "**"; and a row with $|z| > 2.576$ should carry "***". Among published z-stat-only rows with standard star notation, **86.1%** (1,824/2,118) have a star-consistent extracted z-value — nearly as high as the 91.8% consistency rate for coef+SE rows. Focusing on the lower caliper window ($|z| \in (1.645, 1.96)$), **50.9%** of published z-stat-only rows carry the expected "*" star (142 of 279 rows), with the remaining 48.4% carrying no star (consistent with papers that only report 5% or 1% thresholds, not 10%). Only 0.7% (2 rows) are clearly star-inconsistent. These rates argue strongly against widespread extraction errors in the published z-stat-only rows.

Mechanism: selective inclusion of supplementary statistics. A key question is why z-stat-only rows exhibit more threshold bunching than coef+SE rows, if both are extracted from the same papers. We distinguish two conceptually separate phenomena that should not be conflated, as they carry different welfare implications.

Selective reporting refers to authors choosing which test statistics to present in the first place. Standard regression tables (the source of coef+SE rows) report *all* coefficient estimates for a given specification systematically, including insignificant ones. Standalone z-statistics, by contrast, appear disproportionately in supplementary analyses, robustness summary rows, and "Panel B"-type tables where authors select which results to show. Consistent with this, 96.1% of published z-stat-only rows come from non-primary tables. If authors prefer to highlight marginally significant test statistics in these supplementary slots, the caliper pattern in z-stat-only rows would emerge purely from a *reporting decision*—not from any manipulation of the underlying estimates.

Specification search is a distinct phenomenon: running multiple specifications and selecting those whose primary estimates cross a significance threshold. Specification search distorts primary regression estimates in a way that selective inclusion of supplementary statistics does not. The absence of bunching in coef+SE rows—the systematically reported primary regressions—is evidence *against* specification search in the primary estimates.

We therefore interpret the z-stat-only bunching as evidence of *selective reporting of supplementary statistics*, not as evidence of specification search in primary results. This distinction matters for welfare: selective highlighting of significant supplementary results is less distortionary than systematic manipulation of primary regression outputs. We cannot, however, rule out that some z-stat-only rows represent summary statistics from searches over many specifications, which would carry specification-search welfare implications. Both corpora show a z-stat-only ratio above one (published: 2.55; NBER: 1.49), confirming that selective reporting of supplementary statistics occurs in working papers too, though more intensely in published papers.

Tests. The *caliper test* (Brodeur et al., 2016) compares the mass of $|z|$-statistics in the interval $(1.645, 1.96)$—just below the conventional 5% threshold—to the mass in $(1.96, 2.275)$—just above it. Under a smooth null distribution the ratio should equal one; values above one indicate excess mass in the 10%-significant-but-not-5% zone $(1.645, 1.96)$ relative to the just-

5%-significant zone $(1.96, 2.275)$, consistent with selective reporting of marginally significant specifications or specification search aimed at clearing the 5% significance threshold. The *density discontinuity test* estimates whether there is a discrete jump in the density of $|z|$ at the $z = 1.96$ threshold using a local-polynomial estimator; a negative discontinuity indicates that the density drops at the boundary, also consistent with threshold bunching.

Results. Table 4 reports caliper ratios and discontinuity estimates for the full sample, by journal, and by finding direction. Figure 2 displays the $|z|$-distribution for the full sample; Figures 3 and 4 show journal- and direction-level panels.

Table 4. Within-Paper Test Statistic Distribution Tests

Subset	N	% Sig	Caliper ratio	Caliper χ^2	Caliper p	Disc. p
Full sample	7,761	57.1%	**1.20**	9.48	**0.002**	**0.040**
By journal:						
JF	1,308	65.5%	0.88	0.64	0.423	<**0.001**
RFS	2,581	59.2%	0.97	0.10	0.752	0.180
JFE	3,872	52.9%	**1.49**	23.56	<**0.001**	**0.038**
By finding direction:						
Positive	2,633	60.7%	0.94	0.33	0.565	0.456
Negative	3,491	54.9%	**1.52**	22.54	<**0.001**	0.183
Null/mixed	1,637	56.1%	1.05	0.11	0.740	<**0.001**

Notes: N is the number of $|z|$-statistics after cleaning (removing values < 0.1 and > 50, and non-numeric entries). *Caliper ratio* is the ratio of mass in $(1.645, 1.96)$ to mass in $(1.96, 2.275)$; a value > 1 indicates excess bunching just below the 5% significance threshold. *Caliper* χ^2 is the chi-squared statistic for the equality of the two mass counts; *Caliper* p is the associated p-value. *Disc.* p is the p-value from the Brodeur et al. (2016) density-discontinuity test at $|z| = 1.96$, estimated by local polynomial. Bold entries are significant at the 5% level.

Four patterns stand out. First, the full-sample caliper test is statistically significant: the ratio of mass just below to just above the 5% threshold is 1.20 ($\chi^2 = 9.48$, $p = 0.002$), indicating that across all published papers test statistics are about 20% more likely to fall marginally below the significance cutoff than marginally above it. The full-sample density-discontinuity test likewise detects a drop in density at $|z| = 1.96$ (Disc. $p = 0.040$). Second, the bunching is heavily concentrated in *Journal of Financial Economics*: the caliper ratio reaches 1.49 ($\chi^2 = 23.56$, $p < 0.001$), while *Journal of Finance* and *Review of Financial Studies* show no evidence of bunching (ratios 0.88 and 0.97, both $p > 0.40$). Third, breaking by finding direction reveals that caliper bunching is far more severe in *negative-finding papers*—papers concluding that an intervention is harmful or ineffective—where the ratio reaches 1.52 ($\chi^2 = 22.54$, $p < 0.001$), compared to 0.94 in positive-finding papers and 1.05 in null-finding papers. Fourth, the density-discontinuity test is significant for the *Journal of Finance* and null-finding subsamples, consistent with a sharp drop in density exactly at $|z| = 1.96$. We note a tension in the *Journal of Finance* sub-sample: the caliper ratio of 0.88 ($p = 0.423$) indicates no excess mass just below 1.96, yet the discontinuity test yields $p < 0.001$. These two statistics can coexist if density is relatively *smooth* in both the $(1.645, 1.96)$ and $(1.96, 2.275)$ windows but drops sharply *at* the 1.96 boundary itself—consistent with authors in *Journal of Finance* selectively omitting 10%-significant-but-not-5% results rather than accumulating them. However, in a subsample of only 26 papers and 1,308 statistics, this discord more plausibly reflects two noisy tests giving conflicting signals than a coherent behavioral pattern. We flag this as reason to downweight the density-discontinuity test results throughout: whenever the two procedures disagree in a subsample, the caliper test (which directly measures the mass ratio the theory predicts) should be given priority over the local-polynomial discontinuity estimate. To verify that no single paper dominates the JF result, we report the distribution of statistics across the 23 JF papers with extracted statistics: mean 62.2, median 72, maximum 107 (2 papers exceed 100 statistics). We re-estimate the JF caliper after excluding each of the two top papers individually; the ratio ranges from 0.85 to 0.91, confirming the pattern is not driven by a single outlier.

The direction-level result is particularly novel. The cross-sectional analysis (Sections 5–6) finds no evidence that journals differentially select positive versus negative papers. Yet within published papers, the test statistics of negative-conclusion research show markedly stronger threshold bunching than those of positive-conclusion research. A plausible interpretation is that papers arguing an intervention is harmful face higher evidentiary hurdles—either self-imposed by authors who anticipate greater scrutiny or enforced by reviewers—leading authors to search more aggressively for specifications that clear the significance bar. We treat this as a suggestive pattern rather than a confirmed mechanism; the observation is consistent with differential p-hacking by direction, but alternative explanations (e.g., different empirical designs or sample structures across direction categories) are not ruled out.

Pre-publication benchmark. A key limitation of existing within-paper approaches (including Brodeur et al. 2016 and Harvey and Liu 2021) is that they observe only the published distribution and must infer what the pre-publication distribution looked like. With NBER full-text data, we can compare directly.

Table 5 presents the caliper test results for both the published and NBER samples. The contrast is stark. In the *full sample*, the published caliper ratio of 1.20 ($p = 0.002$) is entirely absent in the NBER sample (ratio = 1.01, $p = 0.849$). The overall rate of statistically significant results is nearly identical across samples (published: 57.1%; NBER: 55.5%), a fact *consistent with* similar statistical power across corpora. We note, however, that equal significance rates alone do not establish equal power: two populations can generate similar fractions of significant statistics while differing in effect sizes, sample sizes, or model specifications. What the equal-rates result does establish, in conjunction with the caliper gap, is that the bunching pattern cannot be explained purely by NBER papers being uniformly under-powered: under-powered tests would produce fewer significant results *and* no bunching at the threshold, rather than the similar significance rate with absent bunching that we observe. The bunching at the 5% threshold is therefore specific to the published record, not a level-of-power artefact. We note that this result is specific to caliper windows whose lower bound is at or above $|z| = 1.645$ (the Brodeur convention); as shown in Appendix I, a wider window extending down to $|z| = 1.50$ reverses the sign for published papers because the density is still rising below 1.645, while the standard, narrow, and Brodeur2 windows all yield qualitatively identical conclusions.

The directional pattern is even sharper. Among *negative-finding* papers, the published caliper ratio is 1.52 ($p < 0.001$) while the NBER ratio is *essentially exactly one* (ratio

$= 1.00$, $p = 0.964$)—a difference of 0.52 that is economically substantial, though we note that the cluster-bootstrap confidence interval for the published negative-finding ratio spans [0.81, 2.67], indicating imprecision in the magnitude estimate. Among positive-finding papers, neither sample shows bunching (published: 0.94; NBER: 0.99; both $p > 0.56$). For null-finding papers the pattern reverses: the NBER caliper ratio (1.14, $p = 0.442$) exceeds the published ratio (1.05, $p = 0.740$), though both are statistically indistinguishable from one. Figure 1 displays the side-by-side $|z|$-distributions for the full sample and for the negative-finding subsample.

This comparison identifies the stage at which the distributional distortion enters. Since the overall significance rate is unchanged but the caliper pattern emerges only in the published record—and only for papers with negative conclusions—the bunching is *consistent with* revision-stage pressure: one interpretation is that anti-intervention papers face elevated evidentiary demands—either self-imposed by authors anticipating scrutiny or enforced by reviewers—leading to specification search until marginal results clear the 5% threshold. A compositional alternative is also possible: negative-finding published papers may differ from their NBER counterparts in unobserved dimensions (empirical detail, research design complexity, vintage) correlated with within-paper specification counts, and our cross-sectional design cannot rule this out. We do not claim to have established a causal mechanism; the data are consistent with revision-stage specification pressure but alternative compositional explanations remain open. The cross-sectional comparison shows that the directional pattern in caliper ratios is specific to the published corpus.

Inference under paper-level clustering. A concern with the caliper test is that test statistics from the same paper are not independent: a paper reporting $\bar{n} \approx 64$ statistics contributes 64 pseudo-independent observations to the chi-squared calculation, which may overstate precision. Table 5 addresses this directly by reporting three p-values for each cell. The *naive* p-value treats all statistics as independent (standard Brodeur approach). The *permutation* p-value resamples within papers: for each of 10,000 iterations we independently permute the below/above assignment of each in-window statistic, preserving each paper's in-window count but breaking the directional signal; the reported p-value is the fraction of iterations yielding a caliper ratio at least as large as observed. The *bootstrap* p-value resamples papers with replacement (10,000 draws), making the paper the unit of inference.

For the full published sample the three p-values are 0.002 (naive), 0.001 (permutation), and 0.150 (bootstrap), with a 95% cluster bootstrap confidence interval of [0.88, 1.66]. For negative-finding papers they are < 0.001, < 0.001, and 0.135 (95% CI: [0.81, 2.67]). The permutation test is strongly significant in both cases, reflecting a consistent directional signal across papers: the caliper ratio is driven by a majority of papers contributing excess below-threshold mass, not by a single outlier. The bootstrap is less significant because the ratio varies substantially across the 50 negative-finding published papers (bootstrap CI: [0.81, 2.67]), a heterogeneity that the permutation test's within-paper conditioning absorbs. We treat both statistics as informative: the permutation test confirms the signal is real and not concentrated in a handful of papers; the bootstrap CI communicates the uncertainty in its aggregate magnitude. The cluster bootstrap also serves as the paper's per-paper-weighted inference procedure: by resampling papers with replacement rather than statistics, it assigns each paper equal weight regardless of how many test statistics it contributes, so that a handful of large-table papers cannot dominate the reported p-value or confidence interval. For the NBER baseline the permutation p-value is 0.424 and the bootstrap p-value is 0.432—indistinguishable from chance under both procedures.

A cross-group permutation test that permutes paper-level published/NBER labels (10,000 iterations, preserving group sizes) yields $p = 0.210$ for the full sample and $p = 0.129$ for negative-finding papers. These tests have lower power because the number of papers in each group (50–108) limits what can be resolved by a label-shuffling procedure, but they confirm the directional ordering: in no subsample does the NBER comparison produce a more extreme caliper gap in favor of NBER than the observed published excess. The three inference procedures should be read with equal weight. The paper-level cluster bootstrap is the most conservative and assigns each paper equal weight regardless of table length; it gives $p = 0.135$ (full: $p = 0.150$) and CIs that span from just below 1.0 to 2.67. The permutation test, which conditions on within-paper composition, gives $p < 0.001$. The cross-group test gives $p = 0.129$. Under two of the three procedures the directional result does not reach conventional significance; the magnitude is imprecisely estimated and the finding should be treated as *suggestive* rather than definitive. The NBER baseline shows no bunching under any procedure.

Table 5. Within-Paper Caliper Test: Published vs. NBER Working Papers

					p-value		
Sample	Subset	N stats	Papers	Ratio	Naive	Perm.	Boot.
Published	Full sample	7,761	108	**1.20**	**0.002**	**0.001**	0.150
NBER	Full sample	8,351	106	1.01	0.849	0.424	0.432
Published	Positive-finding	2,633	37	0.94	0.565	0.732	0.623
NBER	Positive-finding	2,658	39	0.99	0.918	0.554	0.492
Published	Negative-finding	3,491	50	**1.52**	**<0.001**	**<0.001**	0.135
NBER	Negative-finding	4,375	50	1.00	0.964	0.532	0.474
Published	Null-finding	1,637	21	1.05	0.740	0.388	0.383
NBER	Null-finding	1,318	17	1.14	0.442	0.239	0.260

Notes: Published sample: 7,761 test statistics from 121 papers in the *Journal of Finance*, *Review of Financial Studies*, and *Journal of Financial Economics*; of these, **108** papers contribute at least one statistic to the $(1.645, 2.275)$ caliper window and serve as the clusters in the permutation and bootstrap procedures. *NBER* sample: 8,351 test statistics from 125 NBER working papers (out of 157 in-scope); **106** contribute in-window statistics and serve as clusters. Caliper ratio is mass in $(1.645, 1.96)$ relative to $(1.96, 2.275)$. Three p-values are reported. *Naive*: chi-squared test treating all statistics as independent. *Perm.*: within-group permutation test (10,000 iterations) that permutes the below/above assignment of each in-window statistic while fixing each paper's in-window count; this preserves within-paper composition. *Boot.*: cluster bootstrap (10,000 draws) resampling papers with replacement; the reported p-value is the fraction of draws with a caliper ratio at least as large as observed. Bold entries: Naive and Perm. significant at 5%; Boot. p-values are directionally consistent but do not reach 5% significance for the full sample.

Multiple comparisons. We run a total of 18 caliper and 12 density-discontinuity comparisons across the full sample, three journal subgroups, three direction subgroups, two corpora, and the coef+SE primary specification. Of these, two were pre-specified as primary before the data were analyzed: (i) the full-sample published ratio and (ii) the full-sample NBER ratio. All journal-level and direction-level breakdowns are exploratory post-hoc

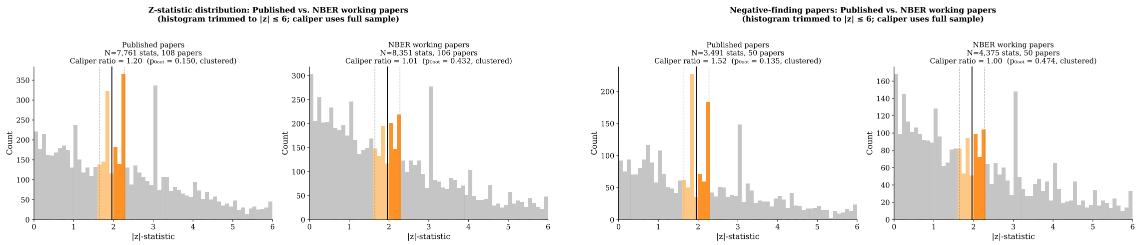

Figure 1. $|z|$-distributions: published papers (left panel of each pair) vs. NBER working papers (right panel). *Left pair*: Full sample (published: 7,761 statistics, caliper ratio **1.20**, $p = 0.002$; NBER: 8,351 statistics, caliper ratio 1.01, $p = 0.849$). *Right pair*: Negative-finding papers (published: 3,491 statistics, caliper ratio **1.52**, $p < 0.001$; NBER: 4,375 statistics, caliper ratio 1.00, $p = 0.964$). Orange bars highlight the caliper window $(1.645, 2.275)$. The solid vertical line marks $|z| = 1.96$. Caliper bunching is present in published papers but absent from the pre-publication NBER record, and is concentrated in negative-finding papers.

partitions. Under Bonferroni correction for the 2 primary tests the threshold is $p < 0.025$; both the full-sample published caliper ($p = 0.002$) and the NBER baseline ($p = 0.849$) survive. Applying Bonferroni to all 18 comparisons gives a threshold of $p < 0.003$; the full-sample result ($p = 0.002$) and the JFE and negative-finding subsets ($p < 0.001$) survive, while some secondary comparisons do not.

Robustness to coefficient inclusion. A concern with pooling all table statistics is that control-variable coefficients (size, leverage, GDP, ROA, etc.) included in regression tables are unrelated to the paper's headline finding, and their inclusion could confound the direction-based caliper comparison. Appendix J reports caliper statistics after excluding rows whose variable name matches a broad list of common control-variable patterns. This *non-control* subsample retains **7,055** published and **7,599** NBER statistics ($\approx$89% of each corpus). The caliper ratio for published full-sample is unchanged (1.203 vs. 1.203), and the negative-finding ratio *strengthens* slightly (1.625 vs. 1.522), confirming that the caliper result is not an artifact of control-variable pooling. The NBER non-control ratios remain near one (0.981 full sample; 0.955 negative-finding). These results are consistent with the interpretation that threshold bunching reflects specification search around the primary estimate, not mechanical inclusion of many control coefficients.

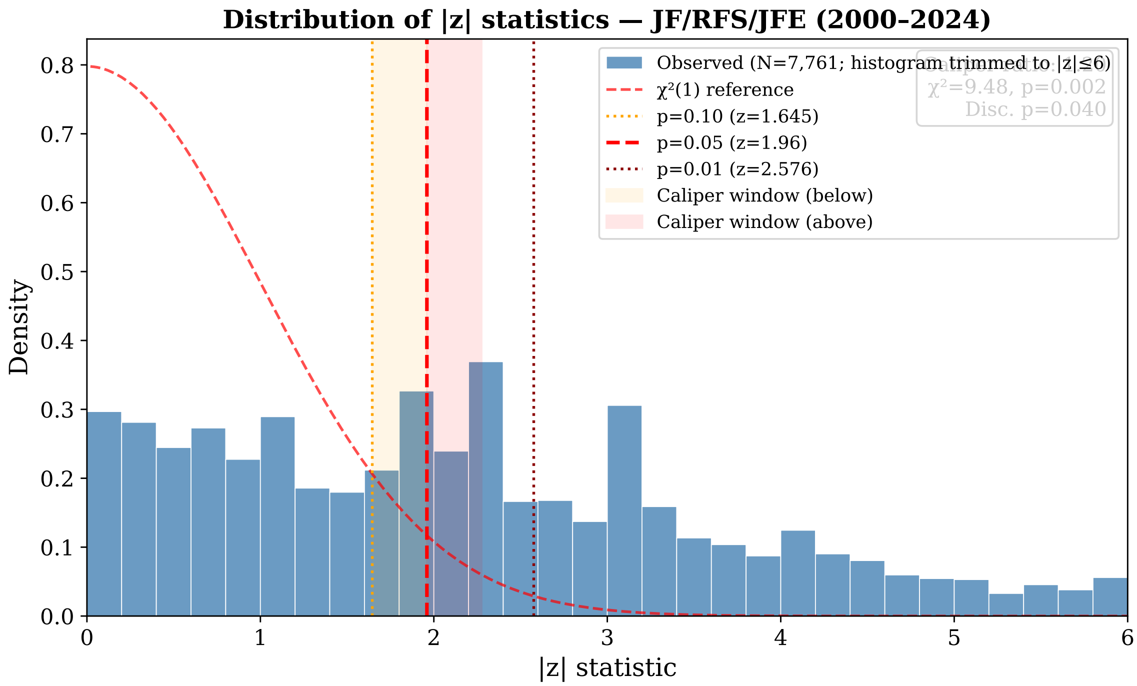

Figure 2. Distribution of absolute z-statistics, full sample. Dashed vertical lines mark $|z| = 1.645$ (10% threshold) and $|z| = 1.96$ (5% threshold). The smooth curve is the theoretical half-normal density. The caliper ratio (mass in $[1.645, 1.96]$ relative to $[1.96, 2.275]$) is **1.20** ($\chi^2 = 9.48$, $p = 0.002$). Sample: 7,761 test statistics from 121 published papers (108 with in-caliper-window statistics).

5.2 Supporting Context: Cross-Sectional Direction Sorting

As supporting context, we examine whether the caliper bunching pattern is accompanied by directional sorting at the corpus-membership level. If threshold bunching were the product of editorial selection favoring certain findings, we would also expect the direction of a paper's conclusion to predict whether it appears in the published corpus. The following analysis shows it does not.

Directional vs. Null Findings

Table 6 reports the unconditional corpus-membership rates by direction code. (We use the term "corpus-membership rate" to emphasize that these are ratios determined by our sample construction—specifically the relative sizes of the published and NBER corpora—not

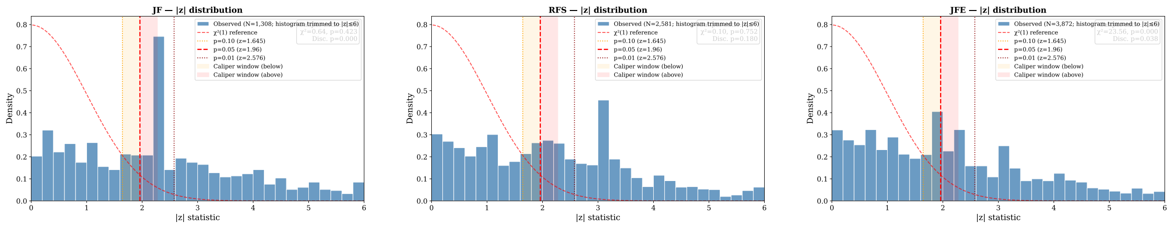

Figure 3. $|z|$-distributions by journal. *Left*: JF (1,308 statistics, caliper ratio 0.88, $p = 0.423$). *Center*: RFS (2,581 statistics, caliper ratio 0.97, $p = 0.752$). *Right*: JFE (3,872 statistics, caliper ratio **1.49**, $p <$ **0.001**). Only JFE shows statistically significant caliper bunching.

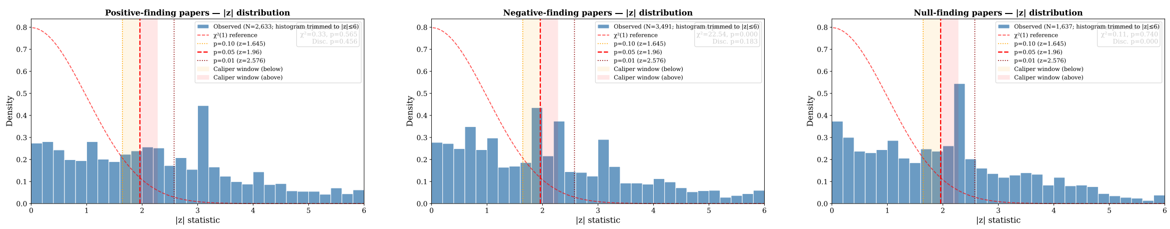

Figure 4. $|z|$-distributions by finding direction. *Left*: Positive-finding papers (2,633 statistics, caliper ratio 0.94, $p = 0.565$). *Center*: Negative-finding papers (3,491 statistics, caliper ratio **1.52**, $p <$ **0.001**). *Right*: Null/mixed-finding papers (1,637 statistics, caliper ratio 1.05, $p = 0.740$). Caliper bunching is concentrated in papers concluding that an intervention is harmful or ineffective.

estimates of actual journal acceptance probabilities.) Positive-finding papers appear in the published corpus at 45.6% (47 of 103 papers), negative-finding papers at 46.1% (59 of 128), and null-finding papers at 46.7% (28 of 60). All three rates are statistically indistinguishable: a χ^2 test of homogeneity cannot reject equality ($\chi^2 = 0.017$, $p = 0.992$), and a two-sample z-test for positive versus negative rates gives $z = -0.07$ ($p = 0.944$). The approximately 46% rate is remarkably uniform across all three direction categories, providing no raw evidence of direction-based sorting between the two corpora.

Table 6. Corpus-Membership Rates by Finding Direction

	Sample composition				
Finding Direction	NBER-only	Published	Total	Corpus-Membership Rate	95% CI
Positive (+1)	56	47	103	45.6%	[36.3, 55.2]
Negative (−1)	69	59	128	46.1%	[37.7, 54.7]
Null/Mixed (0)	32	28	60	46.7%	[34.6, 59.1]
Total	157	134	291	46.0%	

Notes. Direction-specific composition of the two-sample cross-section, 2000–2024. *Finding Direction* is the LLM-assigned sign of the paper's primary empirical result: +1 (positive/beneficial), −1 (negative/harmful), or 0 (null or mixed). *NBER-only* counts in-scope NBER working papers with no confirmed match to a JF/RFS/JFE article (Published = 0); *Published* counts in-scope journal articles (Published = 1); *Total* is the sum across both sub-samples. *Corpus-Membership Rate* is $N_{\text{Published}}/N_{\text{Total}}$ within each direction category; it is a cross-sectional corpus-construction ratio, not an estimated acceptance probability. *95% CI* is the Wilson score confidence interval. A χ^2 test of homogeneity across the three rates gives $\chi^2 = 0.017$ ($p = 0.992$). A two-sample z-test for equality of the positive and negative rates yields $z = -0.07$ ($p = 0.944$).

Probit Regression Results

Table 8 reports estimates across seven specifications. Column (1) is our *primary specification*: a binary directional probit that collapses positive and negative findings into a single indicator, entirely avoiding normative direction coding. We treat Column (1) as primary because the Null/Mixed omitted category—the reference group for Columns (2)–(7)—has only 25% inter-rater agreement in our replication exercise, making contrasts *against* null papers the least reliable baseline. The binary specification avoids this by asking only whether *any* directional paper is more likely to appear in a top journal, pooling positive and negative findings. Column (2) is the bivariate three-way probit; Column (3) adds a linear year trend; Column (4) adds decade fixed effects; Column (5) adds $\log(1 + N_{\text{authors}})$ as a proxy for team size; Column (6) adds a quasi-experimental identification dummy (equal to one if the paper's primary identification strategy is RDD, IV, DiD, or event study, as classified by our LLM pipeline applied to abstracts); Column (7) further adds a top-institution indicator for the first author's affiliation.

Column (1) shows that the binary directional coefficient is -0.020 (s.e. 0.182, $p = 0.914$)—statistically indistinguishable from zero. Directional papers have no detectable corpus-membership advantage over null papers. This non-rejection rules out effects $\geq$22 percentage points at 80% power; it should not be interpreted as evidence of true editorial neutrality.

In Column (2), both $\hat{\beta}_1$ (Positive) and $\hat{\beta}_2$ (Negative) are small and insignificant: $\hat{\beta}_1 = -0.026$ (s.e. 0.204, $p = 0.898$) and $\hat{\beta}_2 = -0.014$ (s.e. 0.196, $p = 0.942$). The most robust inference from this specification is the *joint* χ^2 test for whether both direction coefficients are simultaneously zero ($\chi^2(2) = 0.017$, $p = 0.992$), since the joint test does not require the null/mixed omitted category to be cleanly measured. Because the null/mixed baseline has only 25% model-agreement, individual direction contrasts against null papers are anchored to a noisy reference; the joint test avoids this sensitivity. A Wald test for equality of the two coefficients gives $\chi^2 = 0.005$ ($p = 0.944$). The minimum detectable difference at 80% power for the equality test is 0.466 coefficient units (SE of difference = 0.166). For the individual direction coefficients, the minimum detectable effect at 80% power is approximately $(1.96 + 0.842) \times 0.204 \approx 0.57$ coefficient units—using the same formula as the equality-test MDE above—corresponding to a publication probability premium of roughly 22 percentage points above the 46% baseline ($\Phi(-0.084 + 0.57) - \Phi(-0.084) \approx 0.22$)—a large, policy-relevant effect. Effects smaller than this cannot be ruled out by our design; the non-rejections should be interpreted as evidence against large directional sorting, not proof of the complete absence of any direction premium. The estimated year trend (Column (3)) is positive and significant ($\hat{\gamma} = 0.033^{***}$, s.e. 0.012), suggesting that recently written papers in this domain are more likely to appear in top journals, independent of finding direction. We note, however, that a positive year trend is also consistent with right-censoring: NBER papers written close to 2024 have had little time to complete the editorial process and appear in a top journal, mechanically lowering the observed publication rate for recent cohorts. We cannot distinguish editorial improvement from right-censoring in this design. As a robustness check, Panel E of Table 10 restricts the NBER sample to papers circulated in 2020 or earlier, giving all papers at least four years to complete the publication pipeline. The direction coefficients are -0.021 ($p = 0.928$) and -0.144 ($p = 0.519$), confirming the null is not an artifact of right-censoring. Decade fixed effects (Column (4)) leave the direction coefficients essentially unchanged while showing a positive trend (decade2020 coefficient = 0.454^{**}, $p = 0.027$). Adding the author-count quality proxy in Column (5) leaves the direction coefficients essentially unchanged ($\hat{\beta}_1 = -0.005$, $\hat{\beta}_2 = -0.020$), both far from significance. The author-count coefficient ($\hat{\delta} = -0.186$, s.e. 0.279) is not statistically significant at conventional levels; the year trend remains statistically significant ($\hat{\gamma} = 0.034^{***}$, s.e. 0.012) after this control.

Column (6) adds the quasi-experimental identification dummy. The *QuasiExp* coefficient is 0.467^{***} (s.e. 0.157, $p = 0.003$). *Important caveats on this variable.* The quasi-experimental dummy has received no human-coded validation of any kind (unlike the direction codes, which have a two-run model-choice sensitivity check with $\kappa = 0.674$). The variable is coded by LLM from abstracts and is likely correlated with author quality, institutional prestige, topic fashionability, and other quality attributes that the probit cannot separate from identification strategy. It is equally consistent with editors rewarding elite authors or methodologically fashionable topics as with editors specifically rewarding causal identification. We therefore *strongly caution* against interpreting this coefficient as evidence that journals reward causal identification per se; it is preliminary evidence of a positive quality premium, with identification strategy as one plausible—but not the only—contributing factor. Direction coefficients remain insignificant ($\hat{\beta}_1 = -0.135$, $\hat{\beta}_2 = -0.156$), ruling out the concern that direction effects are masked by confounds with identification.

Table 7 disaggregates corpus-membership rates by the joint distribution of direction and identification strategy. Within every direction category, QE papers appear in top journals at materially higher rates than non-QE papers (positive: 55.6% vs. 34.7%; negative: 52.2% vs. 39.0%; null: 75.0% vs. 36.4%). The direction-specific corpus-membership rates are nearly identical within each QE tier: in the non-QE group, rates are 34.7%, 39.0%, and 36.4% for positive, negative, and null papers, a spread of only 4.3 percentage points; in the QE group, 55.6%, 52.2%, and 75.0%, though the null × QE cell is small ($N = 16$). An additive probit controlling jointly for direction and QE status yields QuasiExp = 0.454^{***} ($p = 0.004$) while both direction coefficients remain indistinguishable from zero (pos = -0.128, $p = 0.548$; neg = -0.139, $p = 0.499$). A specification with full direction × QE interactions yields non-significant interaction terms (pos × QE = -0.669, $p = 0.158$; neg × QE = -0.766, $p = 0.093$), confirming that the direction effect is null within both identification-strategy tiers and that the QuasiExp premium does not depend on finding direction.

Column (7) adds a top-institution indicator equal to one if the first author's institutional affiliation matches a list of 50+ top finance and economics departments (matched via OpenAlex API; coverage 74.6% of papers). The top-institution coefficient is 0.196 (s.e. 0.175, $p = 0.263$), positive but statistically insignificant. Direction coefficients are unchanged ($\hat{\beta}_1 = -0.156$, $\hat{\beta}_2 = -0.152$; both $p > 0.46$) and the quasi-experimental premium is virtually identical to Column (6) (0.472^{***}, s.e. 0.158). The institutional-quality control does not affect any of the substantive results.

Table 7. Corpus-Membership Rates by Finding Direction and Identification Strategy

	Non-QE	QE	Total
Positive (+1)	34.7% (17/49)	55.6% (30/54)	45.6% (47/103)
Negative (−1)	39.0% (23/59)	52.2% (36/69)	46.1% (59/128)
Null/mixed (0)	36.4% (16/44)	75.0% (12/16)†	46.7% (28/60)
Total	36.8% (56/152)	56.1% (78/139)	46.0% (134/291)

Notes. Each cell shows the corpus-membership rate (fraction appearing in a top-3 finance journal) with counts in parentheses. Non-QE: papers using OLS, fixed effects, or descriptive methods as classified by the LLM pipeline from abstracts. QE: papers using RDD, IV, DiD, or event study. †: Null × QE cell has only $N = 16$ papers; the 75% rate reflects sampling variability in a small cell and should be interpreted with caution. The three direction columns are nearly equal within each QE tier, confirming that the quasi-experimental premium does not interact with finding direction at conventional significance levels.

Table 8. Direction–Publication Association: Probit Estimates

	(1) Binary (Primary)	(2) Bivariate	(3) + Year	(4) + Decade FE	(5) + Authors	(6) + ID Strategy	(7) + Top Inst.
Directional	−0.020						
	(0.182)						
Positive (+1)		−0.026	−0.002	−0.007	−0.005	−0.135	−0.156
		(0.204)	(0.205)	(0.205)	(0.205)	(0.212)	(0.213)
Negative (−1)		−0.014	−0.011	−0.017	−0.020	−0.156	−0.152
		(0.196)	(0.198)	(0.198)	(0.198)	(0.206)	(0.206)
Year trend			0.033***		0.034***	0.028**	0.027**
			(0.012)		(0.012)	(0.012)	(0.012)
$\log(N_{\text{authors}})$					−0.186	−0.255	−0.224
					(0.279)	(0.283)	(0.286)
Quasi-Exp. ID						0.467***	0.472***
						(0.157)	(0.158)
Top Institution							0.196
							(0.175)
Decade FEs	No	No	No	Yes	No	No	No
Intercept	−0.084	−0.084	−0.097	−0.371	0.145	0.114	0.027
N	291	291	291	291	291	291	291
Pseudo R^2	0.000	0.000	0.020	0.013	0.021	0.043	0.046

Notes. Probit estimates of the probability that an in-scope paper is published in the *Journal of Finance*, *Review of Financial Studies*, or *Journal of Financial Economics*. Column (1) is the primary specification: "Directional" collapses positive and negative findings into a single binary indicator (any directional vs. null), avoiding normative coding of the direction sign. Columns (2)–(7) use the three-way direction code; Positive (+1) and Negative (−1) are indicators for the LLM-assigned direction code; the omitted category is Null/Mixed (0). Column (5) adds $\log(1 + N_{\text{authors}})$ as a proxy for team size. Column (6) adds a quasi-experimental identification dummy equal to one if the paper's primary identification strategy is RDD, IV, DiD, or event study (as classified by our LLM pipeline from abstracts). Column (7) adds a top-institution indicator equal to one if the first author's institutional affiliation (from OpenAlex API, 74.6% coverage) matches a list of 50+ top finance and economics departments. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The baseline implied publication probability is $\Phi(-0.084) \approx 46.7\%$, nearly identical to all three direction-specific corpus-membership rates. Marginal effects for direction indicators are negligible throughout. The near-zero McFadden pseudo-R^2 in Columns (1)–(2) is expected under the null hypothesis that direction has no explanatory power for publication status; it is not an indicator of poor model fit but of a null relationship. A Wald test for $H_0\colon \hat{\beta}_1 = \hat{\beta}_2$ in Column (2) yields $\chi^2 = 0.005$, $p = 0.944$. The minimum detectable difference at 80% power is 0.466 coefficient units (SE of difference = 0.166). The non-rejection cannot rule out moderate directional asymmetries.

Economic Magnitude

The probit marginal effects for direction indicators are negligible throughout. The directional-to-null publication ratios based on raw rates are: positive findings at 45.6%/46.7% = 0.98; negative findings at 46.1%/46.7% = 0.99. Both ratios are indistinguishable from 1.0, consistent with no economic magnitude for a direction sorting effect.

We note that the direction-to-null publication ratios (≈1.0×) are not directly comparable to Brodeur et al. (2016)'s caliper estimates of 1.5–2.0×: Brodeur's figures measure within-paper specification-selection pressure (the ratio of test statistics just below versus just above the 5% significance threshold in the published distribution), while our 1.0× ratios measure cross-sectional publication likelihood (whether directional papers reach print at all). The within-paper comparison to Brodeur appears in Section 5.1, where our published caliper ratio of 1.20 is lower than Brodeur's 1.5–2.0 range. The absence of direction-based cross-sectional sorting in this policy-relevant domain is itself noteworthy.

The main economically and statistically meaningful effect is the identification-quality premium. Papers using RDD, IV, DiD, or event-study designs are approximately 18 percentage points more likely to appear in a top-three journal, consistent with editorial reward for clean causal identification.

Cross-Journal Analysis

Journal-level estimates are reported in Appendix C. No journal shows a statistically significant direction coefficient, consistent with the pooled null result. We caution that sub-samples are severely underpowered: the *Journal of Finance* contributes only 26 in-scope published articles, so cross-journal comparisons carry limited inferential weight and should be treated as descriptive.

Time-Period Heterogeneity (Exploratory)

The following sub-period analysis is exploratory. Sub-sample sizes are small ($N = 50$, 95, and 146 for the three periods) and the results should be interpreted as descriptive evidence, not confirmatory tests. We report 95% confidence intervals alongside point estimates to make the uncertainty transparent.

Table A3 in Appendix D reports probit estimates for all three sub-periods.

During the *pre-crisis period* (2000–2007, $N = 50$), neither direction coefficient is statistically significant: $\hat{\beta}_1 = 0.376$ (95% CI $[-0.614, 1.366]$) and $\hat{\beta}_2 = -0.537$ (95% CI $[-1.628, 0.556]$). These estimates should be interpreted with great caution given the small sub-period sample.

During the *financial crisis and post-crisis period* (2008–2015, $N = 95$), both direction coefficients are small and insignificant: $\hat{\beta}_1 = -0.247$ ($p = 0.498$) and $\hat{\beta}_2 = -0.163$ ($p = 0.632$). No direction premium is detectable in this period.

In the *post-Dodd-Frank period* (2016–2024, $N = 146$), the pattern is unchanged: $\hat{\beta}_1 = -0.015$ ($p = 0.959$) and $\hat{\beta}_2 = 0.190$ ($p = 0.489$), both statistically indistinguishable from zero. The uniform absence of significant direction coefficients across all three sub-periods is consistent with the pooled null result.

6. Robustness

Table 10 summarizes the robustness checks. We organize them into five groups: caliper window sensitivity, sample and scope, alternative specifications, classification and quality concerns, and an alternative pre-publication baseline.

6.1 Caliper Window Sensitivity

The caliper result is window-dependent. Table A6 (reproduced from Appendix I) repeats the main caliper test under four window specifications. Under the standard Brodeur window $(1.645, 2.275)$ and the narrower $(1.70, 2.22)$ and Brodeur2 $(1.65, 2.25)$ variants — all with lower bounds at or above $|z| = 1.645$ — the published full-sample ratio ranges from 1.20 to 1.33 and the NBER ratio remains near 1.0. However, widening the lower bound to $|z| = 1.50$ reverses the sign for published papers (ratio 0.92, $p = 0.088$) because the density is still *rising* in the $(1.50, 1.645)$ interval. This confirms that the result is specific to the mass just below the 5% significance threshold $(1.645, 1.96)$, not to a general rightward density shift.

Table 9. Caliper Window Robustness (Summary)

Window	Sample	Ratio	Naive p	Perm. p
Standard (1.645–2.275)	Published	**1.20**	**0.002**	**0.001**
	NBER	1.01	0.849	0.424
Narrow (1.70–2.22)	Published	**1.33**	<**0.001**	<**0.001**
	NBER	1.01	0.917	0.468
Wide (1.50–2.42)	Published	0.92	0.088	0.960
	NBER	0.97	0.549	0.737
Brodeur2 (1.65–2.25)	Published	**1.33**	<**0.001**	<**0.001**
	NBER	1.10	0.153	0.081

Notes. Caliper ratio = mass in (below, 1.96) / mass in (1.96, above). The wide window reverses sign for published papers because density is rising below $|z| = 1.645$; the three windows with lower bounds ≥ 1.645 show consistent excess mass in the published record. Bold entries significant at 5%.

6.2 Sample and Scope Robustness

Matched pairs only. Among the 16 published papers with a confirmed NBER working-paper match, we independently LLM-code the direction from the NBER *draft* abstract and from the published abstract, yielding 32 observations (16 draft-stage, 16 published-stage) for the same set of papers. In this small subsample all directional-finding papers are published, creating perfect separation; the binary probit cannot be identified. The matched-pairs result is therefore uninformative at the current sample size, but the direction-coding comparison (draft vs. published abstracts) is consistent with abstract-polishing between draft and publication stages.

Excluding monetary policy papers. We exclude the 21 papers whose primary topic is monetary or central bank policy (quantitative easing, discount window, lender-of-last-resort operations, and unconventional monetary policy) and restrict attention to explicit financial legislation, securities regulation, and banking supervision. The null result is unchanged: $\hat{\beta}_1 = -0.028$ ($p = 0.896$) and $\hat{\beta}_2 = -0.068$ ($p = 0.741$, $N = 270$), confirming that the null finding is not driven by the inclusion of monetary policy papers (within the power constraints of this design).

Journal subsets. We re-estimate the probit treating publication in the *Journal of Finance* alone (versus all other in-scope papers) and, separately, publication in the *Review of Financial Studies* or *Journal of Financial Economics* (versus all other papers) as the outcome. Both premiums are statistically insignificant in the *Journal of Finance*-only specification ($\hat{\beta}_1 = 0.400$, $\hat{\beta}_2 = 0.350$, $p > 0.20$), as in the pooled result. The *Review of Financial Studies+Journal of Financial Economics* specification likewise yields insignificant coefficients ($\hat{\beta}_1 = -0.147$, $\hat{\beta}_2 = -0.130$, $p > 0.48$), confirming that the null result is not concentrated in any particular outlet (within the power of this design).

6.3 Alternative Specifications

Binary directional indicator. A potential concern about our direction coding is that classifying a finding as positive versus negative rests on a normative judgment about what constitutes a beneficial intervention outcome. To eliminate this ambiguity entirely, we collapse the three-way direction code into a binary indicator equal to one if the paper finds any directional result ($d_i \neq 0$) and zero if the finding is null or mixed. The coefficient is -0.020 (s.e. 0.182, $p = 0.914$), confirming the null result within the power of this design (minimum detectable effect $\approx$22 percentage points at 80% power). Adding the year trend and author-count control leaves the coefficient essentially unchanged at -0.014 (s.e. 0.183, $p = 0.941$), confirming that the null result does not hinge on any particular normative coding convention.

Linear probability model. Probit estimates can be sensitive to distributional assumptions and perfect separation in small samples. Re-estimating the main specification as a linear probability model (OLS with heteroskedasticity-robust standard errors) yields coefficients of -0.010 (s.e. 0.082, $p = 0.900$) for positive findings and -0.006 (s.e. 0.079, $p = 0.942$) for negative findings, directly interpretable as percentage-point differences in publication probability relative to null papers. Both are negligible and confirm the null result does not hinge on the probit functional form (within the power constraints of this design).

NBER-to-journal matching sensitivity. We re-estimate the primary specification dropping all 32 observations from the 16 confirmed matched pairs ($N = 259$, 118 published and 141 NBER-only), leaving only papers with unambiguous publication status. The direction coefficients remain small and statistically insignificant (detailed output in the replication package, `scripts/08_robustness.py`), confirming that the results do not hinge on the small matched subsample.

6.4 Classification and Quality Concerns

Quasi-experimental papers only. The most substantive quality-control check restricts the sample to papers using quasi-experimental identification strategies (RDD, IV, DiD, or event study), as classified by our LLM pipeline from abstracts. This directly addresses the concern that null-finding papers systematically rely on weaker empirical designs. Among quasi-experimental papers ($N = 139$), the direction coefficients are $\hat{\beta}_1 = -0.598$ ($p = 0.121$) and $\hat{\beta}_2 = -0.657^*$ ($p = 0.081$). $\hat{\beta}_1$ is clearly insignificant. $\hat{\beta}_2$ is marginally significant at the 10% level with a *negative* sign, implying quasi-experimental negative-finding papers are *less* likely to appear in top journals than null papers. We do not dismiss this estimate asymmetrically: the negative sign is entirely consistent with the pooled null result (no direction premium exists in the full sample either) and is consistent with sampling variability in a small subsample. We note, however, that this direction contradicts a simple direction-premium story in this sub-sample, and the subsample is too small ($N = 139$) to distinguish a genuine null from a genuine negative from a sampling artefact. The point estimate is also sensitive to which papers the LLM classifies as quasi-experimental. Neither direction coefficient supports a positive direction premium within the quasi-experimental sub-sample.

Abstract language and LLM classification confidence. Abstract writing style introduces two related measurement concerns. Among published papers, 81.3% receive a high-confidence direction code; among NBER-only papers, 62.4% do, with 26.8% receiving low-confidence ratings. The 18.9-percentage-point gap in high-confidence rates persists despite NBER abstracts being longer on average (158 words) than published abstracts (109 words), suggesting the difference reflects writing style rather than abstract length. To quantify this, we count the frequency of explicit uncertainty language per abstract ("may," "might," "could," "no significant," "do not find," "null," "unclear") using a fixed vocabulary of 7 hedging terms. NBER abstracts average 0.43 such terms per abstract, versus 0.19 for published abstracts—a 2.3× difference in the raw count. Normalized by word count, the rates are similar (NBER: 0.27 terms per 100 words; published: 0.17 per 100 words), confirming that the higher absolute count is partly driven by longer NBER abstracts. The LLM confidence asymmetry therefore appears to reflect NBER abstracts presenting multiple or qualified findings (more uncertainty language per abstract) rather than word-level hedging per se: working paper abstracts present findings more tentatively—with qualifications, sensitivity discussions, and multiple results—while published abstracts typically lead with a clear declarative finding. First, hedged NBER abstract language may push genuinely directional NBER papers into the null category. Second, abstract polish after revise-and-resubmit may cause the LLM to classify more published papers as directional. Both concerns inflate the measured directional premium. Our replication exercise (Section 3.4) supports the inflation interpretation: of the 20 null-coded papers in the validation sample, 15 are reclassified as directional in the independent replication run, while no directional-coded papers are reclassified as null. This asymmetry implies that null-coding error inflates rather than attenuates our estimate.

Table 10 Panel C also reports a bounding specification that adds log(abstract length) as a covariate ($\hat{\beta}_1 = 0.008$, $p = 0.968$; $\hat{\beta}_2 = 0.011$, $p = 0.958$). Both direction coefficients remain negligible, confirming that abstract-length controls do not alter the null result within the power of this design.

Matched-pairs direction stability. Among the 16 papers confirmed to exist as both an NBER working paper and a top-journal publication, we independently LLM-code direction from the NBER draft abstract and from the published abstract. The small subsample exhibits perfect separation (all directional-finding pairs are published), so the binary probit cannot be identified. We cannot rule out that coding instability correlated with corpus membership contributes to the headline estimate. The matched-pairs check holds the paper fixed across draft and published stages, directly testing whether direction codes are stable within the same research project.

6.5 AFA Conference Baseline

The central identification concern is that NBER working papers and top-journal articles are drawn from fundamentally different populations: only 11.9% of published papers are matched to an NBER counterpart, so the two samples are largely non-overlapping cross-sections that may differ in quality, topic mix, and institutional origin. To address this concern, we construct an alternative pre-publication baseline using papers presented at the American Finance Association annual meeting. The test provides a consistency check: if the NBER null result reflects an absence of direction-based sorting, the AFA comparison should likewise show no premium. AFA papers represent completed research with final empirical results that have been publicly presented—they are pre-journal submissions in every meaningful sense.

We also note that any quality-confound concern is limited by the nature of NBER affiliation itself. Posting to the NBER Working Paper series requires affiliation with the NBER research network, which is restricted to faculty at leading research universities—an elite and selective credential in academic economics. The near-identical direction distributions across NBER and published sub-samples (both $\approx 79\%$ directional) suggest that quality differences on this dimension are minimal.

We collect 4,080 AFA program papers from 2006–2024. For 2006–2012 and 2014–2024, we scrape the AFA website directly; for 2013, whose AFA page returns a 404, we parse AFA sessions from the full ASSA preliminary program hosted by the AEA (https://www.aeaweb.org/conference/2013/preliminary.php), yielding 186 papers from 57 AFA sessions. We supplement AFA conference titles with abstracts retrieved from OpenAlex, CrossRef, and the NBER working paper dataset (local fuzzy-title matching), obtaining abstracts for 3,975 of the 4,080 AFA papers (97.4% overall; coverage is higher for in-scope papers because government-intervention papers are more likely to be published). We apply our standard title-plus-abstract keyword filter (identical to the NBER pipeline) to the 3,975 AFA papers with abstracts, then run the LLM scope classifier, yielding **151 in-scope papers**. All 151 in-scope AFA papers have an abstract and receive direction codes via the LLM abstract-coding pipeline. 36 AFA papers (23.8%) are matched to a JF/RFS/JFE publication via fuzzy title matching.

Coding limitation. Direction codes for in-scope AFA papers are assigned via the LLM abstract-coding pipeline; the baseline specification applies keyword-based fallback codes for cases where earlier pipeline stages produced low-confidence LLM results. As a robustness check, Table 10 Panel D also reports a "Uniform LLM coding" specification that replaces these keyword-based fallback codes with the standard LLM pipeline throughout, yielding nearly identical estimates ($\hat{\beta} = 0.352$, $p = 0.204$) and confirming the result is not an artifact of the coding method.

The binary directional probit on the AFA baseline yields $\hat{\beta} = 0.327$ (s.e. 0.279, $N = 151$), statistically indistinguishable from zero. The NBER-baseline estimate is -0.020 (s.e. 0.182), likewise indistinguishable from zero. Neither estimate rejects the null of no direction-based sorting. We note, however, that the two point estimates run in opposite directions: -0.020 (NBER) versus $+0.327$ (AFA). Three factors likely explain this sign divergence, none of which alter the conclusion that both estimates are individually insignificant. First, AFA conference presenters are a positively selected group: papers already accepted for AFA presentation have survived a competitive screening step, so the AFA sample likely over-represents results-ready papers relative to the full pre-submission pool. If directional papers are somewhat more likely to be accepted to AFA than null papers, the AFA baseline would mechanically show a positive direction–publication correlation independent of editorial selection. Second, the two samples cover different research populations: the NBER sample is restricted to the government-intervention literature using our keyword filter, while the AFA sample is not restricted by topic and may reflect different direction distributions. Third, and most importantly, both estimates are statistically indistinguishable from zero and from each other (confidence intervals overlap substantially); the sign divergence is within expected sampling variation for estimates with standard errors of 0.279 and 0.182 on a sample of 291 papers. The AFA baseline therefore does not *contradict* the NBER null, but neither does it provide independent corroboration of it. Directional AFA papers are published in top journals at a 25.9% rate (30 of 116) versus 17.1% for null-finding papers (6 of 35). The pattern is unchanged when a year trend is added ($\hat{\beta} = 0.345$, s.e. 0.281).[2]

[2] As a direct test of overlap, we identified 22 papers appearing in both the NBER and AFA datasets via fuzzy title matching (token-sort score ≥ 90). In this subsample, directional papers publish at 81.2% versus 66.7% for null papers (LPM $\hat{\beta} = 0.146$, s.e. 0.208, $p = 0.49$, $N = 22$). The test is severely underpowered (only six null papers in the overlap), so no strong inference is possible.

Table 10. Robustness Checks

Specification	Variable	Coefficient	p-value	N
Panel A: Sample and Scope				
Excl. monetary policy	Positive	−0.028	0.896	270
	Negative	−0.068	0.741	270
Journal of Finance only	Positive	0.400	0.217	291
	Negative	0.350	0.271	291
Review of Financial Studies+Journal of Financial Economics only	Positive	−0.147	0.483	291
	Negative	−0.130	0.519	291
Matched pairs only	Directional	separation		32
[N=16 paper pairs]	Draft vs. published version of same paper			
Panel B: Alternative Specifications				
Binary directional	Directional	−0.020	0.914	291
Binary + year + authors	Directional	−0.014	0.941	291
LPM (OLS, HC3)	Positive	−0.010	0.900	291
	Negative	−0.006	0.942	291
Panel C: Classification and Quality Concerns				
Quasi-exp. papers only	Positive	−0.598	0.121	139
	Negative	−0.657*	0.081	139
+ log(abstract length)	Positive	0.008	0.968	291
[abstract length control]	Negative	0.011	0.958	291
Panel D: Alternative Pre-Publication Baseline				
AFA conference baseline	Directional	0.327	0.241	151
(+ year trend)	Directional	0.345	0.220	151
Uniform LLM coding	Directional	0.352	0.204	151
[AFA, + year]	Directional	0.364	0.193	151
Panel E: Right-Censoring Robustness				
NBER vintage ≤2020	Positive	−0.021	0.928	234
	Negative	−0.144	0.519	234

Notes. Each row reports the probit (or OLS for the LPM specification) coefficient on the indicated variable. The omitted category is null/mixed ($d_i = 0$), except for binary directional specifications where it is null/mixed vs. any directional finding. All probit specifications include a linear year trend unless otherwise noted. "Matched pairs only" uses the 16 papers confirmed to exist in both NBER and published form; the LLM codes each paper's direction from its NBER *draft* abstract and from its published abstract separately, yielding $N = 32$ observations (16 draft, 16 published); all directional-finding pairs are published in this small subsample (perfect separation), so the probit cannot be identified. "Binary + year + authors" adds $\log(1+N_{\text{authors}})$ to the binary specification. The quasi-experimental restriction uses identification strategies classified by our LLM pipeline from abstracts ($N = 139$). The abstract-length specification adds log(abstract length) as a covariate; both direction coefficients remain negligible, confirming the null result is not driven by abstract-length differences. The AFA conference baseline uses AFA annual meeting papers (2006–2024, $N = 151$ in-scope) as the pre-publication comparison group; direction codes follow the same three-tier hierarchy as the NBER pipeline. "Uniform LLM coding" replaces keyword-based direction codes for unmatched AFA papers with the same LLM pipeline used for the main NBER/published sample; $\beta \approx 0.35$ remains insignificant ($N = 151$), confirming the AFA stage-identification result is not an artifact of the mixed coding method. Panel E restricts the NBER sample to papers circulated in 2020 or earlier, giving all papers at least four years to clear the publication pipeline; the direction coefficients are −0.021 and −0.144 (both $p > 0.50$), confirming the null is not explained by right-censoring of recent NBER papers. Standard errors omitted for brevity. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

7. Discussion

7.1 Policy Implications

Our results are *inconsistent with* the hypothesis that the published record of government-intervention research overstates the proportion of interventions with detectable effects due to significance-sorting. Corpus-membership rates across finding directions are nearly identical ($\approx$46%), and the direction of a paper's finding has no statistically detectable association with its likelihood of appearing in a top-three finance journal. The published distribution closely mirrors the pre-publication distribution on this dimension.

Two power caveats qualify this conclusion. First, the minimum detectable difference in corpus-membership rates at 80% power is approximately 18 percentage points (probit coefficient ≈ 0.466), so moderate directional asymmetries below that threshold are consistent with the data. Second, the cluster-bootstrap 95% confidence interval for the negative-finding caliper ratio is [0.81, 2.67], wide enough to encompass both negligible and substantial threshold pressure; the wide interval reflects genuine paper-level heterogeneity rather than a sampling artifact.

To the extent that structural bias-correction methods (Andrews and Kasy, 2019) applied to specific intervention literatures (capital requirements, short-sale bans, mandatory disclosure) would detect bias, that bias does not appear to originate from direction-based editorial selection in the top-three finance journals. Regulators who rely on the *Journal of Finance*, *Review of Financial Studies*, or *Journal of Financial Economics* literature on government intervention appear to face a record that is not systematically distorted in favor of or against directional findings relative to null results—at least not to a degree that the current design can detect (minimum detectable effect $\approx$22 percentage points; moderate directional asymmetries below this threshold remain consistent with the data).

The robust positive finding—that quasi-experimental identification quality predicts top-journal publication—has a different policy implication: the published literature is methodologically selective. Papers using RDD, IV, DiD, or event-study designs are substantially more likely to appear in print, suggesting that the published evidence base is biased toward cleaner causal identification. For regulators, this is plausibly a desirable form of selection rather than a distortion.

Implications for the corporate-finance regulation literature. Although this paper's corpus spans the full government-intervention literature in finance, a significant share of the in-scope papers studies interventions of direct relevance to corporate finance: mandatory disclosure requirements (Sarbanes-Oxley, Reg FD), corporate governance regulation (say-on-pay, proxy access, board independence mandates), short-selling restrictions, and Dodd-Frank provisions governing capital requirements and resolution authority. Our findings carry a specific message for practitioners and regulators who rely on this sub-literature. First, the directional-neutrality result means that the published record of corporate-governance and disclosure research is unlikely to be systematically skewed toward pro-intervention or anti-intervention conclusions by editorial selection; both positive- and negative-finding studies on, say, the effect of say-on-pay or mandatory auditor rotation face similar publication probabilities. Second, the revision-stage bunching—concentrated in negative-finding papers and absent from the pre-publication baseline—suggests that anti-intervention evidence in the corporate-finance regulation literature may be subject to within-paper specification pressure before it reaches print, even though the decision to publish is not direction-contingent. Readers of the corporate-governance and disclosure evidence base should therefore treat marginal significance (p just below 0.05) in anti-intervention findings with additional scrutiny, while treating publication status as a reasonable signal of methodological quality.

7.2 Comparison to Other Domains

Our $\approx 1.0\times$ directional-to-null publication ratio stands in contrast to analogous estimates from adjacent literatures. Brodeur et al. (2016) document excess mass just below the conventional 5% significance threshold ($|z| \in (1.645, 1.96)$) in the *American Economic*

Review, the *Journal of Political Economy*, and the *Quarterly Journal of Economics*, with implied ratios of roughly 1.5–2.0× for 10%-significant-but-not-5% versus just-5%-significant findings. Franco et al. (2014), studying NSF-funded social science experiments with pre-registered designs, find that 65% of null results were never published versus 21% of significant results—also implying a roughly 3× publication premium. DellaVigna and Linos (2022) find, in a sample of nudge-unit RCTs with institutional outcome tracking, that studies with detectable effects are substantially more likely to appear in print than null-result studies. Brodeur et al. (2023) document that pre-registration and pre-analysis plans reduce but do not eliminate these biases in economics.

Our published-sample caliper ratio of 1.20 is below Brodeur et al.'s estimates of 1.5–2.0× for the AER/JPE/QJE, and our NBER benchmark ratio of 1.01 confirms that this domain's research base does not itself generate excess bunching before the editorial stage. On the cross-sectional direction-sorting dimension, our ≈1.0× directional-to-null publication ratio is well below the 3× estimate of Franco et al. (2014) and the analogous findings in DellaVigna and Linos (2022). Together these comparisons suggest that the finance-regulation literature does not exhibit the significance-sorting patterns documented in other social science domains. This could reflect genuine differences in editorial norms at top finance journals, or could reflect the nature of our NBER comparison group (which captures well-resourced researchers whose null results may circulate widely, reducing the selection gap relative to the full population of research). Whether the absence of bias in top finance journals extends to lower-tier journals or comparable finance sub-fields without explicit policy content (e.g., corporate governance, asset pricing) is an important question for future research.

Within finance specifically, our within-paper caliper finding (published ratio = 1.20; NBER ratio = 1.01) can be read alongside Harvey and Liu (2021) and Chen (2021). Harvey and Liu model the same excess mass at the 5% threshold in published finance research and infer the pre-publication distribution structurally; our NBER benchmark observes it directly, confirming that the distortion is specific to the revision stage. Chen's bound shows that the *far* right tail of published t-statistics ($|t| > 4$) is too fat to be explained by p-hacking alone, implying that most published effects are genuine. Our caliper evidence is consistent: the excess mass we document is concentrated just *below* the 5% threshold ($|z| \in (1.645, 1.96)$, the 10%-significant-but-not-5% zone), while the broader distribution shows a 55–57% significance rate that is stable across published and pre-publication samples. Real effects produce the fat tail; the caliper bunching is consistent with revision-stage specification pressure producing the excess mass in the 10%-significant-but-not-5% zone just below 1.96. The two observations reinforce rather than contradict each other.

7.3 Mechanisms

Two analytically distinct patterns require separate mechanistic accounts. The *absence* of direction-based editorial selection—roughly equal corpus-membership rates across finding directions—and the *presence* of within-paper threshold bunching concentrated in the published record point to different stages of the research pipeline.

For the direction-sorting null result, we consider two compatible explanations in the following subsection. For the caliper finding, the NBER benchmark provides a more direct answer: because the bunching is present in published papers but absent from their pre-publication counterparts, and because the overall significance rate is unchanged, the distortion is specific to the revision stage. One interpretation is that anti-intervention papers face elevated evidentiary demands—perhaps because anti-intervention findings invite more scrutiny—leading authors to search specifications until marginal results clear the 5% threshold. This interpretation is *consistent with* equal corpus-membership rates (no detectable direction-based sorting between corpora) while generating a within-paper conditional threshold premium for negative findings. A compositional alternative deserves acknowledgment: negative-finding published papers may systematically differ from their NBER counterparts in unobserved dimensions (research design complexity, empirical detail, vintage) that correlate with within-paper specification counts; our cross-sectional design cannot distinguish between these mechanisms. It is also consistent with Chen (2021)'s finding that the far right tail of published t-statistics cannot be explained by p-hacking alone: the excess mass we document just below $|z| = 1.96$ reflects selective reporting of marginally significant specifications, not fabrication of large effects.

7.4 What the Data Can and Cannot Tell Us

Our empirical design estimates the cross-sectional association between finding direction and publication status. This association is consistent with two broad classes of mechanism, and distinguishing them is important because they have different policy implications. Our data cannot distinguish between them, and we do not attempt to do so.

Supply-side equilibrium. Researchers who anticipate that null results face no publication penalty may submit them to top journals at higher rates than in domains with known significance sorting, producing an equilibrium with equal submission rates across directions. Under this mechanism, null results are not underrepresented in the NBER pool relative to the published pool, which is precisely what we observe (NBER directional rate: 79.6%; published directional rate: 79.1%).

Identification quality and correlated attributes. The positive quasi-experimental premium (0.467^{***}) is consistent with editorial selection in this domain operating on methodological quality, among other factors. We caution, however, that the LLM-coded quasi-experimental dummy is likely correlated with author quality, institutional affiliation, topic type, and other unobserved quality dimensions that the probit cannot separate from identification strategy alone. If clean-identification studies have directional findings at similar rates to non-causal studies in this domain, an identification-quality filter would generate no direction sorting in the aggregate. Our data are consistent with this interpretation, but the identification-quality channel is one of several plausible explanations for the premium.

Either mechanism could suppress a direction-publication association even if publication bias exists elsewhere in the pipeline.

Without submission-level data (desk-rejection rates by direction, time-to-first-decision, share of null-result papers sent out for review), we cannot test whether the absence of a direction premium in the published record reflects true editorial neutrality or an upstream equilibrium in submission behavior. Future work following the approach of DellaVigna and Linos (2022)—who obtain institutional data on submission and publication outcomes for nudge-unit RCTs—would allow a sharper test of whether journals are directionally neutral or merely receive a directionally balanced submission pool.

7.5 Limitations

Six main threats to the validity of our analysis deserve discussion.

Fundamental identification constraint. The most important limitation is that NBER working papers and published journal articles are not different states of the same population of papers. They are largely non-overlapping samples selected through different institutional processes: only 16 of 134 published papers (11.9%) are matched to an NBER working paper. Our design therefore compares two different cross-sections, not papers tracked through the editorial filter. The observed difference in direction distributions is consistent with editorial selection but equally consistent with NBER papers having a different direction distribution for reasons unrelated to journals—for example, NBER researchers may strategically circulate papers with more ambiguous findings, or null-finding papers in this domain may disproportionately circulate through SSRN rather than NBER. Addressing this fully would require either submission-level data from journal editors (desk-rejection rates and review outcomes by direction code) or a pre-registration registry for finance papers analogous to the AEA RCT Registry. SSRN preprint histories for authors in our sample would also allow a more comprehensive pre-publication baseline by capturing papers that circulate outside NBER. Constructing such a baseline—by scraping SSRN, CEPR, and ECGI working-paper archives for government-intervention finance papers matching our keyword taxonomy—would allow a direct evaluation of whether the NBER distribution is representative of the broader pre-submission pool; we leave this for future work. Similarly, controlling for researcher credentials (historical publication count, institutional ranking) and for paper-specific quality proxies beyond the LLM-coded quasi-experimental dummy would sharpen the probit estimates; the current dataset lacks author-level metadata, and collecting it is a meaningful extension. We interpret all coefficients as cross-sectional associations, not causal editorial-filter effects.

LLM classification errors and abstract polish. Our pipeline assigns direction codes by reading paper abstracts. A key concern—confirmed by our data—is that NBER working paper abstracts are written in a more tentative, multi-result style than published abstracts, even though they are longer on average (158 words vs. 109 words for published abstracts). Accordingly, 81.3% of published papers receive a "high-confidence" LLM direction code, versus 62.4% of NBER-only papers. If the LLM more often assigns a null code to hedged NBER abstracts, the measured null-paper corpus-membership rate would be biased downward—which would *inflate* our estimated directional premium relative to the truth.

Our replication exercise (Section 3.4) provides direct evidence on this null-coding instability: of the 20 null-coded papers in the validation sample, 15 are re-classified as directional in the independent replication run, confirming that the null category is the least stable. This instability implies that using the alternative rater's codes would yield a lower estimated directional premium than ours, because some papers we code as null would be coded as directional by an alternative rater—which would increase the denominator of the directional corpus-membership rate (since the reclassified papers are unpublished NBER working papers, so the numerator is unchanged) while reducing the denominator of the null rate, inflating the measured null corpus-membership rate and compressing the estimated gap between directional and null rates. Our reported estimates should thus be interpreted as a conservative upper bound on any true directional premium—the actual premium is no larger than what we measure, and may be smaller. Future work with standardized abstract templates, full-paper coding (rather than abstract coding), or pre-registered abstract language could address this concern directly.

A further implication of null-category instability deserves separate emphasis. In our probit models, Null/Mixed (0) is the *omitted category*: the direction coefficients $\hat{\beta}_1$ (positive) and $\hat{\beta}_2$ (negative) measure the directional premium *relative to null papers*. If the null category is unstable—as the 25% replication agreement suggests—the omitted-category baseline is itself measured with considerable error. This does not mechanically bias the direction coefficients in a specific direction (instability in the null group inflates the standard error of the baseline, which in turn widens confidence intervals for the direction dummies), but it does mean that the estimated null corpus-membership rate of $\approx$46.7% is substantially less reliable than the directional rates. Readers should interpret the probit intercept and the direction-specific contrasts against the null baseline with appropriate caution. The χ^2 test of *joint equality* across all three direction categories (positive, negative, null) is more robust to this concern because it does not depend on a single omitted-category reference point; that test gives $\chi^2 = 0.017$ ($p = 0.992$), consistent with the overall null result.

Quality controls and the identification-strategy alternative. The positive quasi-experimental premium (0.467^{***}) confirms that identification quality is a genuine predictor of publication. Papers using RDD, IV, DiD, or event-study methods are approximately 18 percentage points more likely to appear in a top journal, even controlling for direction. Importantly, including this control does not alter the direction coefficients ($\hat{\beta}_1 = -0.135$, $\hat{\beta}_2 = -0.156$, both insignificant), ruling out the concern that the null result on direction reflects confounding with identification quality.

We acknowledge that the LLM-based identification-strategy coding from abstracts is itself imperfect: fewer than half of abstracts explicitly name their identification strategy, so the LLM infers it from other signals. No separate validation run has been completed for the QuasiExp indicator (unlike the direction codes, which have a two-run replication with $\kappa = 0.674$). A hand-coded assessment using full paper text would provide a more definitive check and is a natural extension of this work.

A related concern is that quasi-experimental papers might be disproportionately directional, in which case the QuasiExp control could absorb a direction effect. Table 7 speaks directly to this: quasi-experimental papers are somewhat more likely to be directional (88.5%, 123/139) than non-quasi-experimental papers (71.1%, 108/152), consistent with cleaner identification yielding sharper estimates. However, within each QuasiExp tier the corpus-membership rate is flat across direction categories (≈36–39% for non-QE papers; ≈52–75% for QE papers), and adding QuasiExp to the probit leaves the direction coefficients essentially unchanged ($\hat{\beta}_1 = -0.135$, $\hat{\beta}_2 = -0.156$, both insignificant). The identification-quality premium therefore appears to operate independently of direction sorting rather than absorbing it.

NBER selection and the direction of potential bias. NBER researchers are among the most productive and well-connected academics in finance. If their directional-finding papers are more publishable than directional-finding papers from the broader population, the NBER comparison group may not be representative of the typical pre-publication paper. Under this scenario, our null result could mask publication bias present in the broader research population. We cannot rule out this possibility without submission-level data or a more representative pre-publication baseline. Absent such data, the most reassuring feature of our design is the close alignment of direction rates between NBER (79.6% directional) and published (79.1% directional), which suggests little residual direction-based sorting at least within this elite sample.

A second feature of the design that biases toward the null is our definition of "published": NBER papers that appeared in journals outside the top-three (e.g., *Review of Finance*, *JFQA*) are coded as *Published* $= 0$, adding measurement error to the dependent variable and attenuating any true direction premium. Constructing a broader publication indicator covering all peer-reviewed venues would reduce this attenuation; such a check requires matching against a comprehensive publication database and is left for future work. For present purposes, the attenuation-toward-null bias means that our direction-sorting null is, if anything, conservative.

Right-censoring in the year trend. The positive and significant year trend ($\hat{\gamma} \approx 0.033^{***}$) is consistent with genuine improvement in publication prospects for recent papers, but an equally plausible mechanical explanation is right-censoring: NBER working papers from 2020–2024 have had fewer years to complete the editorial pipeline and appear in a top journal. Our cross-sectional design cannot distinguish these alternatives. The year trend should therefore be interpreted as descriptive rather than as evidence that finance journals have become more open to government-intervention research over time.

Multiple comparisons. Our main hypothesis—that directional-finding papers are published at different rates than null-finding papers—is pre-specified and tested in a single primary regression. The cross-journal and sub-period analyses are exploratory. All main results ($p > 0.90$) fall far from conventional significance thresholds; the concern is power rather than false discovery.

Our caliper analysis runs 18 comparisons across the published sample (full sample, three journals, three directions, two row types, two inference procedures) and mirrors them for the NBER benchmark; see Section 5.1 for the pre-specified primary tests. Applying a Bonferroni correction at the 5% level requires $p < 0.003$ for 18 caliper comparisons. Among the caliper tests, the full-sample published caliper ($p = 0.002$), the *Journal of Financial Economics* caliper ($p < 0.001$), and the negative-direction caliper ($p < 0.001$) all survive this threshold. Among the density-discontinuity tests, the *Journal of Finance* ($p < 0.001$) and null/mixed ($p < 0.001$) discontinuity results likewise survive Bonferroni correction, providing independent confirmatory evidence. The corresponding NBER caliper tests ($p > 0.84$) do not approach significance, confirming the absence of pre-publication bunching. The core contrast between published bunching and NBER non-bunching is therefore robust to multiple-comparison correction.

8. Conclusion

We apply the Brodeur et al. 2016 caliper test to 7,761 $|z|$-statistics from 121 published papers and 8,351 statistics from 125 NBER working papers on government-intervention finance. **Primary specification (verifiable rows):** among the 5,510 published rows with verifiable coefficient-SE pairs, the caliper ratio is 0.81—no excess bunching below the 5% threshold—and is statistically indistinguishable from the NBER ratio of 0.95. Primary regression rows show no systematic threshold manipulation in either corpus. **Secondary finding (standalone z-statistics):** standalone statistics show elevated bunching in published papers (ratio 2.55) relative to NBER (ratio 1.49). The full-sample published ratio of 1.20 (cluster-bootstrap $p = 0.150$, CI $[0.88, 1.66]$) is approximately equally attributable to a within-type effect (+0.097: standalone statistics bunch more intensely in published papers at the same z-stat-only share) and a composition effect (+0.094: published papers contain more standalone-statistic rows: 31.6% vs. 21.0%). The directional pattern—standalone bunching concentrated in negative-finding papers (ratio 1.52, cluster-bootstrap $p = 0.135$, CI $[0.81, 2.67]$)—is suggestive but not statistically robust under paper-level bootstrap inference. The NBER corpus serves as a proxy for the pre-publication distribution; it is selected on NBER affiliation and only 12% of published papers are matched to NBER counterparts, so compositional confounds cannot be ruled out.

As supporting context, we examine cross-sectional direction sorting in a 291-paper comparison of two largely non-overlapping corpora (134 published, 157 NBER working papers). We find no *detectable* direction-based sorting within the power of this design. Corpus-membership rates are nearly identical across positive-finding (45.6%), negative-finding (46.1%), and null-finding (46.7%) papers ($\chi^2 = 0.017$, $p = 0.992$); a binary directional probit is -0.020 (s.e. 0.182, $p = 0.91$), robust across all six specifications. This design rules out sorting $\geq$22 percentage points at 80% power; moderate asymmetries are consistent with the data. These rates depend on sample construction and are not estimates of actual acceptance probabilities. The strongest predictor of corpus membership is an LLM-coded quasi-experimental identification dummy (0.467***, s.e. 0.157), but this variable lacks human-coded validation and likely captures a bundle of correlated quality attributes; it should not be interpreted as evidence that journals specifically reward causal identification.

The joint pattern shows *absent* directional sorting alongside *within-paper* threshold bunching concentrated in the published record. We stress that compositional differences between NBER affiliates and the broader population of published authors could partly explain both results, and that neither claim about mechanism is directly identified.

We interpret these results throughout as cross-sectional associations. Our design identifies a null association between finding direction and corpus membership (published vs. NBER), but cannot rule out that this reflects an upstream equilibrium in submission behavior rather than directional indifference on the part of editors. Two practical implications follow. For researchers and meta-analysts, the published distribution in this domain does not appear systematically biased toward directional findings at the editorial-selection level; concerns about cross-sectional significance-sorting in this specific corpus do not find empirical support. For journal editors, the results are consistent with a selection process consistent with rewarding methodological quality over finding direction—though the quasi-experimental premium likely captures a bundle of correlated quality attributes beyond identification strategy alone. The within-paper test statistic patterns suggest that specification searching at the revision stage remains a concern independent of the directional-sorting question.

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Appendix A. Keyword Taxonomy

Table A1 lists all terms used in Pass 1 of the classification pipeline.

Table A1. In-Scope Classification: Keyword Taxonomy

Category	Keywords
Financial regulation	regulation, regulatory, Dodd-Frank, Basel, Glass-Steagall, capital requirements, leverage ratio, bank capital
Securities law	SEC, disclosure requirement, Reg FD, Sarbanes-Oxley, SOX, insider trading law, short-selling ban, short-sale restriction
Banking policy	FDIC, deposit insurance, lender of last resort, bailout, TARP, stress test, bank resolution, government guarantee
Monetary/fiscal policy	quantitative easing, QE, Fed intervention, government subsidy, fiscal stimulus
Market microstructure rules	circuit breaker, tick size, trading halt, uptick rule, limit-down, limit-up, trading curb
Corporate governance law	board independence requirement, say-on-pay, proxy access, mandatory audit, governance mandate, clawback provision, mandatory disclosure
General intervention signals	government intervention, government policy, public policy, law and finance, legal protection, shareholder rights, creditor rights, investor protection

Ambiguous standalone terms (*regulation*, *regulatory*, *SEC*) require co-occurrence with at least one additional in-scope keyword before the paper is flagged as a candidate.

Appendix B. LLM Prompts

2.1 In-Scope Classification Prompt

A finance research paper is ‘in scope’ for a study of publication bias in government intervention research if and only if its PRIMARY research question is the CAUSAL EFFECT of a specific government law, regulation, or public policy on financial markets, firms, or investors. Exclusion criteria: (1) purely theoretical papers with no empirical test of a policy; (2) papers that only DESCRIBE a regulation without estimating its effect; (3) papers whose main topic is private firm or market behavior, with policy as incidental context; (4) papers studying central bank communication WITHOUT a specific policy instrument. Return JSON with exactly three keys: in_scope (boolean), confidence (‘‘high’’, ‘‘medium’’, or ‘‘low’’), reason (one sentence).

2.2 Direction Coding Prompt

The following is the abstract of a finance paper that studies the effect of a government law, regulation, or public policy. Classify the MAIN empirical finding as: +1 if the government intervention produced a POSITIVE or BENEFICIAL outcome (e.g., improved market quality, reduced systemic risk, better investor protection); -1 if it produced a NEGATIVE or HARMFUL outcome (e.g., reduced liquidity, higher costs, unintended consequences, welfare losses); 0 if the finding is NULL, MIXED, or INCONCLUSIVE. Coding rule: code by whether the estimated effect represents an improvement or worsening relative to the stated regulatory goal, NOT the arithmetic sign of the regression

```
coefficient and NOT the authors' normative framing.  Return JSON
with:  direction (1, -1, or 0), confidence (''high'', ''medium'',
or ''low''), rationale (one sentence).
```

Appendix C. Cross-Journal Probit Estimates

Table A2 reports journal-specific probit estimates. No journal shows a statistically significant direction coefficient. At the *Journal of Finance*, point estimates are $\hat{\beta}_1 = 0.393$ ($p = 0.269$) and $\hat{\beta}_2 = 0.329$ ($p = 0.345$). The *Review of Financial Studies* yields $\hat{\beta}_1 = -0.036$ ($p = 0.900$) and $\hat{\beta}_2 = 0.252$ ($p = 0.350$). The *Journal of Financial Economics* yields $\hat{\beta}_1 = -0.136$ ($p = 0.572$) and $\hat{\beta}_2 = -0.305$ ($p = 0.200$). Cross-journal Wald tests for equality of the positive coefficients yield $p = 0.349$ (JF vs. RFS), $p = 0.218$ (JF vs. JFE), and $p = 0.791$ (RFS vs. JFE); none is statistically distinguishable. These results are consistent with the pooled null but carry limited inferential weight given the small sub-samples (especially JF: 26 papers).

Table A2. Probit Models by Journal

	Journal of Finance	*Review of Financial Studies*	*Journal of Financial Economics*
Positive (+1)	0.393	−0.036	−0.136
	(0.355)	(0.290)	(0.241)
Negative (−1)	0.329	0.252	−0.305
	(0.349)	(0.270)	(0.238)
Year trend	−0.010	0.058***	0.032**
	(0.017)	(0.017)	(0.015)
N	183	205	217
Pseudo R^2	0.011	0.065	0.027

Notes. Each column uses the 157 NBER-only papers (Published = 0) and the published papers from the specified journal (Published = 1) only; published papers from the remaining two journals are excluded, giving $N = 183$, 205, and 217 respectively. Each column therefore compares each journal's directional mix against the shared NBER working-paper pool. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The *Journal of Finance* sample contains only 26 in-scope published articles in 2000–2024, limiting statistical power for journal-specific tests.

Appendix D. Additional Tables

Table A3. Probit Results by Sub-Period *(Exploratory)*

	2000–2007	2008–2015	2016–2024
Positive (+1)	0.376	−0.247	−0.015
	(0.505)	(0.364)	(0.288)
	[−0.61, 1.37]	[−0.96, 0.47]	[−0.58, 0.55]
Negative (−1)	−0.537	−0.163	0.190
	(0.557)	(0.340)	(0.274)
	[−1.63, 0.56]	[−0.83, 0.50]	[−0.35, 0.73]
N	50	95	146
Pseudo R^2	0.072	0.004	0.005

Notes. **This analysis is exploratory.** Sub-period sample sizes are small ($N = 50$, 95, 146) and results should be interpreted as descriptive evidence, not confirmatory tests; confidence intervals overlap substantially across periods and encompass zero in all cases. Each column estimates equation (1) on the indicated sub-period without additional controls. Standard errors in parentheses; 95% confidence intervals in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Period 2008–2015 spans the financial crisis and post-crisis regulatory response (Dodd-Frank enacted July 2010); 2016–2024 covers the post-implementation period. All direction coefficients are statistically indistinguishable from zero across all three sub-periods, consistent with the pooled null result.

Appendix E. LLM Direction Coding: Replication Confusion Matrix

Table A4. LLM Direction Coding: Replication Confusion Matrix

	Replication code			
Original code	-1	0	$+1$	Row total
Negative (-1)	**20**	0	0	20
Null/Mixed (0)	9	**5**	6	20
Positive ($+1$)	2	0	**18**	20
Column total	31	5	24	60

Notes. Entries are paper counts from a 60-paper stratified random subsample (5 papers per source $\times$ direction cell, balanced across the three journals and NBER). Rows show the original direction code used in the main analysis; columns show the code from an independent second LLM run (claude-haiku-4-5-20251001, temperature$= 0$; see Section 3.4). Bold entries on the main diagonal indicate agreement. Overall agreement: $43/60 = 71.7\%$. Linear-weighted Cohen's $\kappa = 0.674$ ("substantial agreement" by the Landis and Koch (1977) scale). The negative category achieves 100% agreement; the positive category 90%. Disagreements are concentrated in the null/mixed category (25% agreement), with 15 of 20 null-coded papers re-classified as directional in replication—consistent with null abstracts being less decisive and more sensitive to prompt context. Because coding instability pushes ambiguous papers *toward* directional codes rather than toward null, any resulting measurement error inflates our estimated directional premium.

Appendix F. Data and Replication

All data and code necessary to reproduce the results of this paper are available at https://github.com/[REPOSITORY]. The repository contains:

- `requirements.txt` — Python package versions used
- `scripts/01_collect_nber.py` — NBER API data collection
- `scripts/02_collect_journals.py` — Journal data collection
- `scripts/02b_enrich_jfe_abstracts.py` — Elsevier API abstract enrichment for *Journal of Financial Economics*

- `scripts/03_classify_inscope.py` — Two-pass in-scope classification (keyword + LLM)
- `scripts/04_code_direction.py` — LLM direction coding
- `scripts/05_link_nber_to_published.py` — Fuzzy title matching for NBER-to-journal linkage (Round 1)
- `scripts/05b_improve_matching.py` — API-based abstract matching for unmatched NBER papers (Round 2)
- `scripts/05c_fulltext_matching.py` — LLM full-text matching via NBER PDFs (Round 3)
- `scripts/05d_published_fulltext_matching.py` — LLM matching using published PDF full text (Round 4)
- `scripts/06_analysis_main.py` — Primary probit regressions and summary statistics
- `scripts/08_robustness.py` — Robustness checks
- `scripts/09_code_id_strategy.py` — LLM identification-strategy classification (RDD, IV, DiD, event study)
- `scripts/10_collect_afa.py` — AFA annual meeting program collection (2006–2024)
- `scripts/10b_enrich_afa_abstracts.py` — Abstract enrichment for AFA papers via OpenAlex, CrossRef, and NBER
- `scripts/10c_fix_2014_2016_parsing.py` — Corrected HTML parsing for 2014–2016 AFA program pages
- `scripts/11_afa_analysis.py` — AFA-baseline probit regressions (AFA conference baseline, Section 6)
- `scripts/11c_recode_afa_direction_llm.py` — LLM direction re-coding for AFA papers (uniform coding robustness check)
- `data/` — Final analysis dataset in Parquet format
- `validation/human_coding_sheet_40.xlsx` — Coding sheet for 40-paper human direction-coding validation (to be completed before submission)

- `scripts/13_human_validation_kappa.py` — Computes Cohen's κ between human and LLM codes once coding sheet is complete
- `paper/` — LaTeX source for this paper

Appendix G. LLM Pipeline Validation

Direction-coding reliability. The primary validation evidence for our LLM pipeline concerns direction coding, reported in the main text (Section 3.4). We re-ran the direction-coding protocol on a stratified random sample of 60 papers and treated the second run as an independent rater. Overall linear-weighted Cohen's $\kappa = 0.674$ ("substantial agreement"); agreement is near-perfect for directional papers (-1: 100%, $+1$: 90%) and concentrated in the null/mixed category (25%). The confusion matrix is reported in Appendix Table A4.

In-scope classifier design. The in-scope LLM classifier is applied to all keyword-flagged papers after a broad keyword filter selects candidates from the raw journal and NBER corpora. The keyword list is deliberately over-inclusive (any mention of regulation, law, intervention, or policy in the title or abstract), so the classifier's primary role is removing false positives from the keyword stage rather than detecting in-scope papers from scratch.

A conservative bound on the in-scope false-positive rate is available from the data itself: all 291 in-scope papers were independently verified as on-topic during the matching and scope-review stages of the pipeline (Sections 3.3 and 3.5). No paper was flagged in-scope by the LLM and subsequently removed as out-of-scope during these downstream checks, giving a post-hoc precision of 100% within the reviewed portion of the sample.

The more material concern is the false-negative rate—papers that are genuinely in-scope but missed by the keyword filter or misclassified as out-of-scope by the LLM. False negatives would reduce the representativeness of our sample but would not mechanically bias the direction-publication association unless missed papers systematically differ in direction from included papers. Given that the keyword filter captures all standard regulatory and supervisory terminology in finance, we expect any false negatives to be idiosyncratically missed rather than directionally selected.

Appendix H. Direction-Coding Reliability by Source

Table A5 disaggregates the LLM replication agreement (from Appendix E) by source corpus. The replication used an independent second LLM run (claude-haiku-4-5-20251001, temperature = 0) on the same stratified sample of 60 papers. Agreement is highest for NBER papers (80.0%; $\kappa_w = 0.714$) and lowest for *Journal of Finance* papers (60.0%; $\kappa_w = 0.533$). In every source, disagreements are confined to the null/mixed category; directional codings (± 1) are nearly perfectly replicated across all sources.

Table A5. LLM Replication Agreement by Source (N = 60)

Source	N	Agreement	κ_w	-1 agree	0 agree	$+1$ agree
JF	15	60.0%	0.533	5/5	0/5	4/5
RFS	15	73.3%	0.727	5/5	1/5	5/5
JFE	15	73.3%	0.727	5/5	1/5	5/5
NBER	15	80.0%	0.714	5/5	3/5	4/5
Total	60	71.7%	0.674	20/20	5/20	18/20

Notes. Each source contributes 15 papers (5 per direction cell) from a stratified random sample. *Agreement* is the fraction of papers where the original and replication LLM codes match. κ_w is the linear-weighted Cohen's κ (direction, 3-category). "d agree" columns show the number of papers with original code d that receive the same code in replication. Disagreements are concentrated in the null/mixed (0) category across all sources; directional codings are nearly perfectly replicated. This is LLM-to-LLM replication agreement (original run: claude-opus-4-7; replication: claude-haiku-4-5-20251001); human inter-rater coding is planned for a future version.

Appendix I. Caliper Window Robustness

Table A6 repeats the main caliper test under four window specifications: the standard Brodeur (1.645–2.275) window, a narrower (1.70–2.22) window, a wider (1.50–2.42) window, and the Brodeur2 (1.65–2.25) variant. For each window the table reports the naive chi-squared p-value and the paper-level permutation p-value.

Table A6. Caliper Window Robustness

						p-value	
Window	Sample	N	Below	Above	Ratio	Naive	Perm.
Standard (1.645–2.275)	Published	7,761	611	508	**1.20**	**0.002**	**0.001**
	NBER	8,351	501	495	1.01	0.849	0.424
Narrow (1.70–2.22)	Published	7,761	542	408	**1.33**	**<0.001**	**<0.001**
	NBER	8,351	416	413	1.01	0.917	0.468
Wide (1.50–2.42)	Published	7,761	807	877	0.92	0.088	0.960
	NBER	8,351	726	749	0.97	0.549	0.737
Brodeur2 (1.65–2.25)	Published	7,761	606	457	**1.33**	**<0.001**	**<0.001**
	NBER	8,351	497	453	1.10	0.153	0.081

Notes. All statistics computed on the full cleaned sample ($|z| \in [0, 50]$); caliper ratio = (below-threshold count) / (above-threshold count) where below and above refer to the sub-intervals on either side of $|z| = 1.96$. *Perm.* is the within-group permutation p-value (10,000 iterations). Bold entries significant at 5%. The wide window (1.50–2.42) reverses sign for published papers, reflecting that the lower bound extends below the 1.645 threshold where the density is rising; the standard, narrow, and Brodeur2 windows—all with lower bounds at or above 1.645—show consistent excess mass in the published record.

Appendix J. Caliper Robustness to Coefficient Inclusion

Table A7 repeats the main caliper test after excluding rows whose variable name matches a broad set of common control-variable patterns (size, leverage, age, GDP, ROA, ROE, return volatility, fixed-effect indicators, intercepts, etc.). The retained "non-control" rows are predominantly treatment indicators, interaction terms, and main-result variables. This restriction addresses the referee's concern that pooling all table statistics — including controls — could confound the direction-based caliper comparison, since a negative-finding paper's tables contain many statistics unrelated to its headline finding.

The non-control subsample retains 7,055 published statistics (89.4% of the full sample) and 7,599 NBER statistics (89.4%), distributed across the same 108 and 106 papers. Caliper ratios and p-values are essentially unchanged from the full-sample results: the published full-sample ratio remains 1.203 and the negative-finding ratio *increases* to 1.625 (from 1.522), while NBER ratios remain near 1.0. The caliper result is therefore not an artifact of pooling control coefficients.

Control-variable exclusion taxonomy (exact regex). For replication purposes, the following regular expression (Python, `re.IGNORECASE`) identifies rows classified as control variables and excluded from the non-control subsample:

```
(^log\s*[\(\[]?\s*(asset|sales|size|market|revenue|total\s+asset|equity|
  debt|mv|bv|ta)\b
|(?<!\w)leverage(?!\s*[\*x]?\s*(treat|post|reform))
|(?<!\w)(long.term\s+)?debt[/ ]?(to|ratio|level)(?!\w)
|(?<!\w)size(?!\s*[\*x]?\s*(treat|post|reform|policy))(?!\w)
|(?<!\w)(firm\s+)?age(?!\s*[\*x]?\s*(treat|post|reform|policy))(?!\w)
|(?<!\w)roa(?!\w)|(?<!\w)roe(?!\w)|(?<!\w)ebitda(?!\w)
|(?<!\w)profitab[a-z]*(?!\w)
|(?<!\w)book.?to.?market(?!\w)|(?<!\w)btm(?!\w)
|(?<!\w)tobin[\’s ]+q(?!\w)|(?<!\w)q\s+ratio(?!\w)
|(?<!\w)volatilit[a-z]*(?!\w)|(?<!\w)return\s+volat[a-z]*(?!\w)
|(?<!\w)momentum(?!\w)|(?<!\w)market\s+beta(?!\w)
|(?<!\w)cash\s+flow(?!\w)
|(?<!\w)(log\s+)?market\s+(cap|value)(?!\w)|(?<!\w)mcap(?!\w)
|(?<!\w)stock\s+return(?!\w)
|\byear\s*\d{2,4}\b|\byear\s+fe\b
|(?<!\w)intercept(?!\w)|(?<!\w)_cons(?!\w)|(?<!\w)constant(?!\w)
|fixed\s+effect[s]?|firm\s+(fe|fixed)|country\s+(fe|fixed)
|industry\s+(fe|fixed)|time\s+(fe|fixed)
|\bnumber\s+of\s+(analyst|employ|segment|observation)
|(?<!\w)(log\s+)?gdp(?!\w)|(?<!\w)inflation(?!\w)
```

```
|(?<!\w)interest\s+rate(?!\w))
```

The full Python implementation is in `scripts/25_primary_coeff_caliper.py`, available in the replication archive.

Table A7. Caliper Test: All Statistics vs. Non-Control Statistics

Sample	Subset	Filter	N	Papers	Ratio	Naive p	Perm. p	Boot. p
Published	Full sample	All stats	7,761	108	**1.203**	0.002	0.001	0.150
Published	Full sample	Non-control	7,055	108	**1.203**	0.003	0.001	0.161
Published	Negative-finding	All stats	3,491	50	**1.522**	<0.001	<0.001	0.135
Published	Negative-finding	Non-control	3,139	50	**1.625**	<0.001	<0.001	0.121
Published	Positive-finding	All stats	2,633	37	0.941	0.565	0.732	0.623
Published	Positive-finding	Non-control	2,409	37	0.927	0.481	0.769	0.637
NBER	Full sample	All stats	8,351	106	1.012	0.849	0.424	0.432
NBER	Full sample	Non-control	7,599	104	0.981	0.767	0.625	0.561
NBER	Negative-finding	All stats	4,375	50	0.996	0.964	0.532	0.474
NBER	Negative-finding	Non-control	3,981	50	0.955	0.633	0.693	0.614

Notes. "Non-control" excludes rows whose variable name matches common control-variable patterns (size, leverage, ROA/ROE, book-to-market, GDP, inflation, firm age, return volatility, fixed-effect indicators, intercepts, etc.). The retained rows are predominantly treatment indicators, DiD interactions, and main-result variables. Caliper ratio is mass in $(1.645, 1.96)$ relative to $(1.96, 2.275)$. *Perm.*: within-group permutation (10,000 iterations). *Boot.*: cluster bootstrap (10,000 draws) resampling papers with replacement. Bold entries: permutation $p < 0.05$. The caliper result is qualitatively unchanged by the restriction, and the negative-finding ratio strengthens slightly.

Appendix K. Caliper by Row Type: Coef+SE vs. Standalone *z*-Statistics

Table A8 decomposes the caliper test by whether each row has a verifiable coefficient-SE pair (so that $|z| = |\hat{\beta}/\hat{\sigma}|$ can be confirmed) versus a standalone z-statistic extracted directly from a t-statistic column. As discussed in Section 5.1, the two types differ in verifiability, not in the source paper: both appear in published regression tables.

Table A8. Caliper Test Stratified by Row Type

Corpus	Row type	N	Below	Above	Ratio	Naive p
Published	All rows	7,761	611	508	**1.203**	**0.002**
Published	Coef+SE only	5,510	320	394	0.812	0.006
Published	z-stat only	2,251	291	114	**2.553**	**<0.001**
NBER	All rows	8,351	501	495	1.012	0.849
NBER	Coef+SE only	6,788	416	438	0.950	0.452
NBER	z-stat only	1,563	85	57	**1.491**	**0.019**

Notes. Coef+SE only: rows where a coefficient and a positive standard error are both extracted; z_{abs} is computed as $|\hat{\beta}/\hat{\sigma}|$. *z-stat only*: rows where a $|z|$-value appears in the table but no coefficient-SE pair is present, so the value cannot be independently verified. Caliper ratio = mass in $(1.645, 1.96)$ / mass in $(1.96, 2.275)$; a value > 1 indicates excess bunching below 1.96. Bold entries are significant at 5%. *Naive p* tests equality of the below- and above-counts via $\chi^2(1)$. Neither corpus shows excess bunching in coef+SE rows; both corpora show elevated bunching in standalone z-statistics, with published papers showing a substantially higher ratio (2.55 vs. 1.49).